OWNER : **INDIAN OIL CORPORATION LIMITED (IOCL)**

PMC : **TOYO ENGINEERING INDIA PRIVATE LIMITED (TEIPL)**

LSTK CONTRACTOR : **L&T HYDROCARBON ENGINEERING LIMITED (LTHE)**

UNIT : **COMMON / 002**

IOCL/ TOYO JOB NO. : **6389 / IPEG**

LTHE JOB NO. : **IOCL-O&U**

Drawing/Document Category (Submission Purpose)	**Drawing/Document Review Status (By OWNER / PMC)**		
	Approval Code	**Status**	**Action Required**
☑ For Approval	1	Rejected	Revise & resubmit
☐ For Review	2	Commented as noted	Work may proceed subject to acceptance & incorporation of comments
☐ For Information	3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final. Work may proceed.
	4	Approved with No comments-FINAL	Work may proceed.
	5	Information Only	Accepted for information only. Work may proceed.

Rev.	Description	Prepared By	Reviewed By	Approved By	Reviewed By	Approved By	Approved Date
A1	Issued for Approval	APG	HPP / YUK	CAM	-	RMK	29.03.2019

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REVISION HISTORY SHEET

S. No.	Rev. No.	Page No.	Clause / Line No.	Brief Description / Reason of Revision	Remarks
1	A1	-	-	Issued for Approval	-

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ANNEXURE -1 SITE SPECIFIC SEISMIC SPECTRA……..69

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1. GENERAL

1.1 Scope

This document covers the Engineering design basis / philosophy to be used for Civil, Structural, Building & Architectural works for the Ethylene Glycol production facility comprising of an ERU (060 - Ethylene Recovery Unit), EGU (061 - Ethylene Glycol Unit), Cooling Tower Unit (127A), Fore water booster pump house for EGU & ERU (127C), Ethylene storage sphere & BOG package (207A), Ethylene Glycol (EG) Storage tanks (207B), Tanker loading & unloading area (207C), TEG Barreling facility (207D), OSBL pipe / Utility facilities & buildings at Paradip Refinery Complex at IOCL, Dist. Jagatsinghpur, Orissa, India.

Paradip refinery is located at Paradip on the east coast of India in the State of Odisha, at a distance of approx. 5 Km from Paradip Port. The site is about 90 Km by road from Cuttack and about 120 Km from Bhubaneswar, the Capital of Odisha, which is well connected by road, rail & air with other parts of the country.

Brief scope of work for General Civil, Structural, Building & Architectural works shall be, but not limited to the items specified, as under for this project:

Major items for General Civil works:

- UG Facilities : Storm Water Drainage, CRWS network, OWS network
- UG RC Structure : Vessel Pit, Valve Pit, Manhole, Catch Basin
- Trenches : Cable Trench & Pipe Trench
- Roads, Pavement and RCC Culverts
- Foundation For :

1) Vertical Equipment : Process Columns & Vessels
2) Horizontal Equipment : Exchangers, Vessels & Drums
3) Package Skid i.e Lube Oil, Doasing Package, Chemical Injection Package etc
4) Block foundation for Pump & Blowers
5) Tank & Sphere
    - Isolated / Combined / Thickened Pave Foundation for miscellaneous supports for EL/IN/PP, Pipe Sleepers etc

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Major items for RC / Steel Structure works:

- Structural Steel Pipe rack (with/without, Air Fin Cooler & Header Structures)
- Equipment Supporting Structures in Steel
- RC Table Top Structures supporting EO Reactor
- Compressor Shelters, Shed for various plant facilities as indicated in plot plan
- Cooling Tower Basin & Structure
- Operating Structure & Platforms
- Equipment Mounted Platforms
- Fire proofing / Painting of Structural Steel as per specification & fire proofing envelop as defined in hazardous area classification drawing in compliance to applicable codes & standards.

Building works:

- Main Sub-Station Building SS-315 N for ERU & EGU
- Sub-Station Building SS-316 N for Cooling Tower
- Sub-Station Building SS-317N for BOOT – 3 area
- Satellite Rack Room SRR 822 for ERU & EGU
- Satellite Rack Room SRR 824 for Cooling Tower
- Satellite Rack Room SRR 827 for BOOT – 3 area
- CT Chemical house (430)
- S & D Building (428)
- Sample Retention room (429)
- Car parking & ladies / gents toilet
- Extension of PP unit Lab building
- Control room / operator room (207C) loading / unloading gantry area
- Fire water booster pump house / shed

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1.2 Standard Definitions

Following standard definitions shall be applied.

Owner	Indian Oil Corporation Limited (IOCL) ; Paradip Refinery
Licensor	CB&I Technology Inc.(for ERU) Scientific Design (for EG)
PMC	TOYO Engineering India Pvt Ltd
EPC Contractor	L&T Hydrocarbon Engineering (LTHE); has been awarded for LSTK-1 (EG-ERU Package) & LSTK-2 (Offsite & Utility Package)
Detail Engineering Contractor (DEC)	L&T – Chiyoda Ltd.

Note: LTHE (as EPC Contractor) has been awarded both the packages i.e. LSTK-1 (EG – ERU) & LSTK-2 (Offsite & Utility). Hence, this design philosophy has been prepared as a common document for LTHE scope of work, which shall be followed for both the packages for synergy and uniformity in design approach.

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2. SPECIFICATIONS, CODES AND STANDARDS

All analysis / design shall be based on the project specific design basis prepared by DEC in compliance project / standard specifications as received in Tender & latest edition of applicable codes and standard. Intent of this design basis is to provide brief extract of the basic design to be followed in conjunction with the tender specification & the latest versions of relevant Indian standards and codes.

In case of any difference in requirements between various specifications including tender specifications and the codes / standards, difference shall be brought to the notice of Client / PMC for necessary clarification and consideration in design meeting client's requirement.

2.1 List of Applicable Indian Codes & standards

The analysis and design of all Structures and Buildings shall be in accordance with, but not limited to, the following BIS codes and standards:

Loading	IS-875 (Part 1 to 5)	Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures
	IS-1893 (Part 1) (2016) IS-1893 (Part 4): (2015)	- Criteria for Earthquake resistant design of Structures - Criteria for Earthquake Resistant Design of Structures. Part 4 Industrial Structures including Stack-like Structures
Concrete Design and Detailing	IS-456 (2000)	Plain and Reinforced Concrete – Code of Practice
	IS-13920 (2016)	Ductile Detailing of Reinforced Concrete Structures subjected to Seismic Forces – Code of Practice
	IS 2911	Design and construction of Pile Foundation
	IS 2974	Code of Practice for Design and Construction of Machine foundations; Part 1 to 5
	IS 3370	Code of Practice for Concrete Structures for Storage of Liquids; Part 1 to 4
	IS 6403 (1981)	Code of practice for determination of bearing capacity of shallow foundation
	SP 16	Design Aids for Reinforced Concrete to IS 456:

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		1978
	SP 34	Handbook on Concrete Reinforcement and Detailing
	SP 20	Handbook on Masonry Design and Construction
	IS: 4326 (2013)	Criteria for Earthquake Resistant Design & Construction of Buildings
Steel Design	IS: 800 (2007)	General construction in Steel – Code of practice
	IS: 808 (1989)	Dimensions for Hot Rolled Steel beam, column, channel & Angle sections.
	IS: 2062 (2011)	Steel for general structural purpose.

Permissible Stresses shall be as specified in the codes mentioned below:

a. Structural members

IS: 800	Hot rolled sections (excluding transmission towers and switchyard structures)
IS: 801	Cold formed light gauge sections
IS: 806	Tubular structures

b. Bolts

IS: 800	Hot rolled sections (excluding transmission towers and switchyard structures)
IS: 801	Cold formed light gauge sections

c. Welds

IS: 816	Metal Arc Welding
IS: 801	Cold formed light gauge sections

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2.2 **List of Project Specific Documents (Available in Tender)**

Sr. No.	Description	Document No.
1	Basic Engineering Design Basis	PDRP-8820-SP-0001 (Rev F2)
2	Notes To LSTK Contractor (For MEG Unit)	IEGP-6351-8110-PH-000-0014 (Rev.T1)
3	Notes To LSTK Contractor For ERU	IEGP-6351-8110-PH-000-0016 (Rev.T1)
4	Scope Of Work General	IEGP-6351-8840-TN-000-0001 (Rev T1)
5	RA Methodology & Assumptions IEGP	IEGP-6351-8110-RA-000-0001
6	Civil scope of work for (LSTK-1)	IEGP-6351-8310-TN-000-8007 (Rev T2)
7	Civil scope of work for (LSTK-2)	IEGP-6351-8310-TN-000-8008 (Rev T3)
8	Construction Scope Of Work IEGP	A – 5.3
9	Foundation Design Basis IEGP	IEGP-6351-8310-PH-000-8001 (Rev.T2)
10	Detail Engineering requirements for Civil, Structural & Architectural works	IEGP-6351-8310-SP-000-8002 (Rev T1)
11	Specific Design requirements	IEGP-6351-8310-SP-000-8003 (Rev T1)
12	Structural Steel Work Design	PDRP-8320-SP-0016 (Rev F6)
13	Civil Design Load & Load Combination	IEGP-6351-8310-SP-000-8006 (Rev T1)
14	Design Loading For Buildings, Process Structures, Pipe Racks and Equipment Foundation	PDRP-8310-SP-0002 (Rev.F2)
15	Specification For Wind And Earthquake Loading	PDRP-8310-SP-0001 (Rev.F1)
16	Architectural Specification For Buildings	IEGP-6351-8310-SP-000-8004 (Rev T1)
17	Concrete Foundation, Structure & Paving	PDRP-8310-SP-0007 (Rev F4)
18	Anchor Bolt And Fixing Standard	PDRP-8310-SP-0003 (Rev.F2)

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19	Concrete Fire Protection	PDRP-8310-SP-0011 (Rev.F2)
20	Passive Fire Protection & External Fireproofing Of Equipment Supports & Structures	PDRP-8440-SP-0007

Note: Above listed are the main design guideline, which shall be referring to various other tender guidelines, specification, design basis, standard drawings etc. LTHE shall be considering all such relevant tender documents for design purpose, though are not listed in table above.

2.3 List of Standard Drawings & Document

Project specific standard drawings prepared by DEC shall be used as first preference for the job. Standard drawings which are not covered by DEC shall be referred from following IOCL's list of project specific standard drawings.

IOCL – Standard Drawings (from Tender)

Sr. No.	Description	Drawing No.	Sheets
1	Standard Details of Manhole, Catch Basin	PDRP-8310-45-0002 (Rev S2)	1
2	Detail of roadway	PDRP-8310-45-0006 (Rev S1)	1
3	Storm water details	PDRP-8310-45-0003 (Rev S2)	1
4	Sanitary Sewer Lift station and Manhole Details	PDRP-8310-45-0008 (Rev S2)	1
5	Concrete Pavement, KERB, Joint and Minor Foundation Paving	PDRP-8310-45-0009 (Rev S2)	1
6	Cable trench & duct bank electrical details	PDRP-8310-45-0007 (Rev S2)	1
7	Standard steel works notes and details	PDRP4200-8320-SP-1001 (Rev A1)	1
8	Fencing details	PDRP-8310-45-0010 (Rev S1)	1
9	Standard Steel Work Notes And Details	PDRP4200-8320-SP-1001 (Rev.A1)	53 Page

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Note: Above listed are the main design guideline, which shall be referring to various other tender documentation. LTHE shall be considering all such relevant tender document for design & construction purpose, though are not listed in table above.

2.4 List of Conceptual Architectural Drawings

Conceptual Architectural drawings received as part of tender document shall form a basis for DEC to develop the detail building layouts with necessary updation and modification as needed to fulfill the project requirement.

Architectural requirement i.e. Finish Schedule, Door Window sizes and type etc shall be in compliance to the conceptual architectural drawings duly approved by IOCL during tender stage.

Requirement other than above, not specified in the conceptual architectural drawing shall be referred from applicable project specific specifications, standards & drawings.

Conceptual Architectural Drawings:

Sr. No.	Description	Drawing No.	Sheets
1	Main Sub-Station Building SS-315 N for ERU & EGU	IEGP-6351-8310-47-062-8106 to 8108	3
2	Sub-Station Building SS-316 N for Cooling Tower	IEGP-6351-8310-47-062-8101, 8102 & 8109	3
3	Sub-Station Building SS-317N for BOOT – 3 area	IEGP-6351-8310-47-062-8110 8112	3
4	Satellite Rack Room SRR 822 for ERU & EGU	IEGP-6351-8310-47-062-8103 to 8105	3
5	Satellite Rack Room SRR 824 for Cooling Tower	IEGP-6351-8310-47-062-8116 to 8118	3
6	Satellite Rack Room SRR 827 for BOOT – 3 area	IEGP-6351-8310-47-062-8113 to 8115	3
7	S & D Building (428)	Annexure 2 of Doc. No. IEGP-6351-8310-TN-000-8008	1
8	Sample Retention room (429)	Annexure 1 of Doc. No. IEGP-6351-8310-TN-000-8008	1

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9	Extension of PP unit Lab building	IEGP-6351-8310-47-062-8125 & 8126	2
10	CT Chemical house (430)	Footprint / conceptual layout shall be as per unit plot plan. However, same shall be updated by LTHE as per detail engineering requirement.	-
11	Car parking & ladies / gents toilet	Same as above	-
12	Control room / operator room (207C) loading / unloading	Same as above	-

3. UNITS OF MEASUREMENT

Units shall be in accordance with the SI system for all activities concerning Civil, Structural Steel & Building design.

Weight	:	kN
Length	:	meter, millimeter
Area	:	Sq. M
Volume	:	Cu. M
Temperature	:	°C

4. BASIC ENGINEERING DESIGN DATA

Following data shall be considered for design:

a. **Rainfall intensity**: (Ref. Cl.3.4 of Doc. No. PDRP-8820-SP-0001)
The maximum annual rainfall is 2251.7 mm is distributed throughout the year from the month of May to November. For design, the rainfall intensity shall be considered as 125 mm/hour in line with tender specifications.

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b. **Wind intensity**: (Ref. Cl.4.3.1 of Doc. No. PDRP-8310-SP-0001)
In line with the above stated tender specification, basic wind speed for design shall be considered as ~65 m/s (i.e. 55 m/s basic wind speed x 1.15 cyclonic zone factor = 63.25 m/s Say 65 m/s conservatively).

c. **Seismic Zone:** (Ref. Cl. No. 6.5 "Results" of "Final Report Earthquake Engineering Analyses Paradip Refinery Project, Orissa, India. Project No. 3193.026, Sept'2008)
Referring to conclusion drawing in above referred tender document, the site specific spectra developed in Sept'2008 for Paradip complex is higher than the codal spectra as given IS 1893 (Part-4) 2015 for "Zone-III".

LTHE has prepared the comparison of both the above stated spectra and the most conservative one has been adopted for early start of design activity by LTHE. Ref. Annexure-1 for the M/s FUGRO's Sept'2008 Vs. IS 1893 Spectra.

Further as per tender requirement, LTHE is conducting separate study for site specific spectra of LSTK-1 & LSTK-2 scope of work.

LTHE shall update this section upon finalization of spectra as stated above. Also necessary design verification as needed will be performed for structures / foundation issued for construction based on M/s FUGRO's Sept'2008 spectra report.

d. **High Point of Paving (HPP)**:

In compliance to the requirement as specified in following listed tender documents;
Doc.No. IEGP-6351-8230-01-000-0001_Rev.T1 "Overall Plot Plan Ethylene Glycol (IEGP)" ;
Doc.No. IEGP-6351-8310-45-000-8201_Rev.T1 "Civil Scope Of Work For Unit 060 ERU, 061 EG, SRR-822 & SS-315 N (LSTK-1)";

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High Point of Paving is considered as under for LSTK-1 Scope of work area;

Unit Area Mark	**HPP Elevation**
Crown of new road	It should be at EL + 100.550 (RL +4.610)
ERU (060)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro grading HP El + 100.000 (RL +4.060) Sloping towards peripheral drain beyond RC paved area
EG (061)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro grading HP El + 100.000 (RL +4.060) Sloping towards peripheral drain beyond RC paved area
Sub-Station (SS 315N)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro Grading - HP/FGL EL +100.000 (RL +4.060) sloping towards peripheral drain for future SubStation area
Satellite Rack Room (SRR 822)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro Grading - HP/FGL EL +100.000 (RL +4.060) sloping towards peripheral drain for future Satellite Rack Room area

In compliance to the requirement as specified in following listed tender documents;
Doc.No. IEGP-6351-8230-01-000-0001_Rev.T1 "Overall Plot Plan Ethylene Glycol (IEGP)";

Doc.No. IEGP-6351-8310-45-000-8202_Rev.T2;

Doc.No. IEGP-6351-8310-45-000-8203 Rev.T2 "Civil Scope Of Work For LSTK-2";

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High Point of Paving is considered as under for LSTK-2 Scope of work area;

Unit Area Mark	**HPP Elevation**
Crown of new road	It should be at EL + 100.550 (RL +4.610)
Ethylene storage Spear Area (207A)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro grading HP El +100.000 (RL + 4.060) sloping towards peripheral drain beyond RC paved area & under offsite rack
Firewater booster pump house / shed (127C)	- RC Paved HPP/FGL EL +100.300 (RL +4.060) with in shed ; 900mm wide Plinth Protection around building - Micro grading HP El +100.000 (RL + 4.060) sloping towards peripheral drain beyond plinth protection
Offsite Racks	Micro Grading FGL EL +100.000 (RL +4.060)
Cooling Tower Area (127A)	- RC Paved Area : HPP EL +100.300 (RL +4.360) - Micro grading HP El +100.000 (RL + 4.060) sloping towards peripheral drain beyond RC paved area & under offsite rack
Cooling Tower : Sub-Station (SS 316N)	Micro Grading - HP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around building sloping towards peripheral drain
Cooling Tower : Satellite Rack Room (SRR 824)	Micro Grading - HP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around building sloping towards peripheral drain
PP Lab Building Extension	- Building Plinth level FFL +100.300 (RL + 4.360) + 900mm wide Plinth Protection around building - Micro Grading - HPP/FGL EL +100.000 (RL +4.060) ; sloping towards peripheral drain
TEG Barreling Unit (207D)	Micro Grading - HPP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around Barreling facility
BOOT3 Sub-Station (SS 317N) & BOOT3 Satellite Rack Room (SRR-827)	Micro Grading - HPP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around Buildings sloping towards peripheral drain

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EG Storage Tank Area (207B)	RC Paved Area : HPP EL +100.000 (RL +4.060) within dyke area and pump shed area Micro Grading - HP/FGL EL +100.000 (RL +4.060) sloping towards peripheral drain
EG Gantry Area (207C)	RC Paved Area : HPP EL +100.000 (RL +4.060) Micro Grading - HP/FGL EL +100.000 (RL +4.060) sloping towards peripheral drain
Sample retention room	Micro Grading - HPP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around Buildings sloping towards peripheral drain
S & D Building	Micro Grading - HPP/FGL EL +100.000 (RL +4.060) ; 900mm wide Plinth Protection around Buildings sloping towards peripheral drain

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5. MATERIALS OF CONSTRUCTION

Material of construction for Major Tags / Items shall be in line with the following:

Structure	Material Of Construction
Pipe Rack with Air-Fan Cooler "PR"	Structural Steel + Grating floor under Air-Fin Cooler & PSV floor. Fireproofing up to Air Fin Cooler Support Elevation (SOW)
Pipe Rack w/o Air-Fin Cooler "PR"	Structural Steel + Grating Floor as applicable for PSV floors / access / walkways (Fireproofing as applicable based on OISD 164)
Equipment Supporting Structure "ES" & Operating Platforms "OS"	Structural Steel + Grating Floor as per requirement (Fireproofing as applicable based on OISD 164)
Compressor Shelter	Structural Steel + Grating Floor as per operation requirement around compressor Roof Sheeting : - 0.5mm thk GI Pre-coated profile sheet - Intermittent skylight panel of clear embossed polycarbonate profile sheet (2mm thk., both side UV stabilized, 60%~80% transparency) Side Cladding: - Shall be 0.5mm thk GI Pre-coated Profile sheet. Side cladding shall start from 3m above HPP elevation and shall extend up to roof with provision of louvers for ventilation. Gantry girder for EOT crane shall have 1m wide walkway on one side (long travel) with handrail on both sides.

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Pump Sheds	Structural Steel Roof Sheeting : - 0.5mm thk GI Pre-coated profile sheet - Intermittent skylight panel of clear embossed polycarbonate profile sheet (2mm thk., both side UV stabilized, 60%~80% transparency) Side Cladding: - Shall be 0.5mm thk GI Pre-coated Profile sheet. Side cladding shall be hung from roof to provide adequate rain protection to the Pumps inside.
EO Reactor Structure	RC Table Top Structure upto EL +121.000 supporting EO Reactor on RC deck slab. Structure above EL +121.000m shall be in Structural Steel as required with grating floor. Operating floors below EL +121.000m shall be in Structure Steel with Grating floor
Horizontal Exchanger & Drums (Grade Mounted)	RC foundations with raised pedestal shall support the equipment as per process / piping requirement. Operating platform around equipment, as needed shall be in structural steel as per plant & piping requirement.
Compressor Foundation	RC Table Top / RC Block Foundation as per vendor drawing requirement
Pump & Package Foundation (Skid Mounted on Grade)	RC Table Top / RC Block Foundation as per vendor drawing requirement
Plant & Non Plant Buildings	Sub-Station Buildings (SS315N, SS316N, SS317N) :- RC Frame Structure + Brick Infill Wall + 0.5mm thk. Profile GI Pre-coated Roof Sheeting over Transformer Bay Satellite Rack Room (SRR822, SRR824, SRR 827), S & D Building, CT Chemical House, Sample Retention Room, Extension of PP lab building : RC Frame Structure + Brick Infill Wall

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Satellite Rack Room (SRR822)	RC Blast Resistant Structure for Blast Loading as per Cl. 5.0 of PDRP-8310-SP-0004
Cooling Tower	RC Basin + RC Super Structure with (Cast-in-situ / precast fins as per vendor details) + Steel Stair etc as per the requirement specified by vendor

5.1 Structural Concrete

Minimum concrete grade, unless otherwise specified in the design drawings, shall be as stated here under in compliance to tender specification Table-3 of Doc. No. PDRP-8310-SP-0005 & provision of IS 456: 2000 (Environmental Condition "Sewer for structural use of concrete").

Description	Grade Designation (Design mix)	Min Cement Content kg/m^3	Max Water/Cement Ratio
Cast in Situ concrete for Piles	M30	400 Note-1	0.45
Reinforced concrete for all general usage including Paving, Manhole, Catch basin, Valve pit, Drains, Trenches, Foundations, RC Structure-Buildings etc but not limited too	M30	320	0.45
Unreinforced Concrete : Blinding Concrete / PCC under RC Foundation, Pits, Trenches etc	M15	240	0.60
Mass Fill Concrete : Where ever required for general structural fill purpose to fill loose soil pocket / under pump / block foundation etc	M10 (1:3:6) Nominal Mix	240	0.50

Note-1 : Ref. Cl. 7.7.1 of ES-AZ 312 I IN "Construction Specification For Bored Cast In-situ Concrete Piles", "With proper mix design and used of proper admixture the cement content may be reduced to but in no case the cement content shall be less than 350 kg/m^3 ".

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Plain Cement Concrete (PCC) shall be in line with the following table as per Cl. No. 7.2.6 of Doc. No. PDRP-8310-SP-0007.

Description	Conc. Grade	Thickness (mm)	Side Projection (mm)
Under all Pile Cap	M15	75	50
Under all Shallow Foundation	M15	75	50
Under Foundation for liquid retaining structures.	M15	100	75
Under Masonry wall Foundation	M20	150	50

5.2 Cement

Ref. to Cl. 2.4 of Doc. No. IEGP-6351-8310-PH-000-8001_Rev.T2, in compliance to Tender Geotechnical investigation report, indicates sub-soil/sub-soil water as non-aggressive in nature.

Accordingly, usage of Ordinary Portland cement (OPC) & Portland Pozzolana Cement (PPC) is recommended in compliance to IS-269 / IS-8112 / IS 12269 / IS-1489 / IS 455 as applicable.

Ref. Table-3 of Doc. No. PDRP-8310-SP-0005 "General Specification for Concrete Works", Cement shall be of Grade-53 for Reinforced Concrete Works, whereas Grade-43 may be used for Unreinforced Concrete Works such as; blinding concrete under foundation & mass filling works.
NOTE: 43 grade (OPC/PPC) cement should be used for structural concrete / PCC / lean concrete subjected to prior approval from PMC / Owner.

5.3 Reinforcing Steel Bar

Ref. to Cl. 5.8 of Doc. No. PDRP-8310-SP-0007_Rev.F1, High strength deformed TMT bars of grade Fe500D / Fe550D, having minimum elongation of 14.5% conforming to IS: 1786 & IS: 13920 shall be used as reinforcement for all reinforced concrete works.

18 gauge black soft annealed SWG wire shall be used for binding of reinforcement bars.

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Minimum size of reinforcement bar shall be used for various element shall be as under;

- Piles & Footing : 12mm
- Columns : Main Bar - 12mm; Stirrups & Ties – 8mm
- Beams : Main Bar - 10mm; Stirrups - 8mm
- Wall : Main Bar - 8mm; Distribution Bar – 8mm
- Structural Slabs : Main Bar - 8mm; Distribution Bars – 8mm
- Non Structural Grade Slab : Main Bar - 6mm; Distribution Bars – 6mm
- Paving (Light Duty) : 100mm thk: 6mm dia @ 200c/c wire mesh (Cl. 8.1 of Doc. No. PDRP-8310-SP-0013)
- Paving (Heavy Duty) : 150mm thk: 6mm dia @ 200c/c wire mesh in Tow layers i.e Top & Bottom (Cl. 8.1 of Doc. No. PDRP-8310-SP-0013)
- Misc. RC Element : 8mm
- Block Foundation : 8mm skin reinforcement (main & distribution)

5.4 Aggregates

All aggregates shall conform to IS: 383 and IS: 515, coarse aggregate shall be crushed from blue basalt/hard granite and shall be hard and free from weathering. The material shall be graded to achieve homogeneous mix sourced from approved vendors by PMC / Owner. Size of aggregate shall be in compliance to Table-3 of Doc. No. PDRP-8310-SP-0005 "General Specification For Concrete Works"

Fine Aggregate: This shall be free from dust and silty sand and from original impurities. River sand from sources / vendor approved by PMC/Owner shall only be used for Concrete works.

The grading of fine aggregates shall be determined as per IS: 2386 (Part I) and shall be within the limits given in Table 4 of IS: 383. Sand shall also have to be washed off if it contains silt more than allowable limits.

5.5 Admixtures

Various admixtures used for improving concrete workability, water tightness shall be used as per vendor's recommendation as needed especially for large pours, liquid retaining structures etc, with prior approval from person in-charge Owner / PMC at site as per Cl. 4.9 of Doc. No. PDRP-8310-SP-0005.

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5.6 **Water Stop**

Proprietary PVC or Hydrophilic, swellable, elastomeric strip material (150mm wide x 5mm thk) shall be used as water stop for all liquid retaining structures such as Pits, Catch Basin, Manhole.

Above requirement is not applicable to Cable / Pipe Trenches & "U" ditches as are not intended to have water tightness as liquid retaining structures.

5.7 **Grout**

Grout material & its application shall be in line with the following requirement:

Type	Application	Grout Material Specification	Grout Thickness
G1	Under Base Plate of ; - Pipe Racks, Technological Structures, Operating Structures, Pipe Supports & Stanchions etc - Static Equipment i.e. Exchangers, Vessels, Drums, Process Columns, Transformers etc - Compressor, Pumps & Blowers (unless otherwise specified by the vendor as Epoxy Grout) - Package Skids i.e. Injection Package, Lube Oil Skid etc.	A proprietary non-shrink, non-metallic, non-corrosive grout which shall be installed in accordance with the Manufacturer's written instructions.	30 mm (min) or As per vendor's recommendation with min. compressive strength of 40 MPa
G2	Under Base Plate of ; Compressor, Pumps, Blower etc as recommended by vendor / machine supplier	A proprietary epoxy grout which shall be mixed and installed in accordance with the manufacturer's written instructions.	30 mm (min) or As per vendor's recommendation
G3	Under Base Plate of ; Stair, Ladder, Misc. Pipe Supports, EL-IN Junction Boxes etc where it is required only for levelling purpose.	Sand-cement grout. The compressive strength of the grout shall not be less than 14 N/mm^2 and 22 N/mm^2 at 7 days and 28 days respectively. Mix shall be 1:1 (sand :	25mm (min)

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		cement) for bedding thicknesses not greater than 25mm and 2:1 for bedding thicknesses not greater than 50mm.	

(Note : If required by vendor, base frames core shall be filled with M15 Grade PCC to augment the mass)

5.8 **Masonry Works** (Ref. Cl. 8.2 of Doc.No. PDRP-8330-SP-0020)

Bricks for masonry works shall be class 5.0 conforming to IS:1077 in compliance to.

External Walls : Shall be 230mm thk in 1:6 cement / sand mortar

Internal Partition Wall : Half brick (115mm thk), in 1:4 cement / sand mortar with two number of 6mm diameter mild steel bars provided at every fourth course.

Fire Wall – Transformer Compound : shall be 200mm RCC Or 355mm thk Brick masonry

5.9 **Structural Steel**

(Ref. PDRP-8320-SP-0016 "Structural Steel Work Design")
(Ref. PDRP-8320-SP-0017 "Structural Steel Work Fabrication")
(Ref. PDRP-4200-8320-SP-1001 "Standard Steel Work Notes and Details"

Structural steel work design, fabrication, construction & material shall be in compliance to the requirement specified in above listed tender specification & in accordance with IS: 800-2007.

Item Description	Grade	Material Code
Hot Rolled Section for Structural Use (Main Frame Members i.e Columns & Beams)	E250B UTS - 410 MPa YSt - 250 / 240 / 230 MPa	IS: 2062-2011 ; IS 808 & IS 800
Section for Structural Use as Minor Platforms, Handrail, Ladders, Stairways etc ; (Tertiary beams, joist, Channels, Angle, Plates, Flats, Rods etc)	E250B preferred (Alternatively E250A is acceptable as per tender specification)	IS: 2062-2011 ; IS 808 & IS 800

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Hollow Sections (RHS / SHS)	UTS - 410 MPa YSt - 250 / 240 / 230 MPa (Higher grade as manufactured by supplier may be used as per design needs)	IS 1161 ; IS 4926 ; IS 800
Tubular Section For Structural use	UTS - 410 MPa YSt - 250 / 240 / 230 MPa	IS 1161-1998 ; IS 1387 ; IS 800
Cold Rolled Sections For Structural use as ; Purlin, Sheeting etc.	As per manufacturer detail	IS 513 & IS 811
Chequered plate (6mm min thk on plain)	E250A/B (as per supplier spec)	IS 3502 Material : IS 2062
Steel Grating	Bearing Bar : E250A/B Twisted Cross Bar as per supplier specification	Shall confirm to IS: 226 Material : IS 2062
Rails	UTS - 410 MPa YSt - 250 / 240 / 230 MPa	Shall confirm to IS: 3443

Finish should be in compliance to PDRP-8440-SP-0006 "Protective Paint And Coatings".

5.10 **Steel Connections**

(Ref. PDRP-8320-SP-0016 "Structural Steel Work Design")
(Ref. PDRP-8320-SP-0017 "Structural Steel Work Fabrication")
(Ref. PDRP-4200-8320-SP-1001 "Standard Steel Work Notes and Details"

a) All structural field connections shall be bolted connections unless noted otherwise in the engineering drawing as welded connection especially.
b) Shop connections shall be welded connection as per engineering drawing.
c) Minimum size of weld shall be 6mm & 8mm respectively for Shop weld & Field weld.
d) Plan bracings are considered as bolted connection irrespective of structure is welded or bolted type.
e) Bolted connections shall be as under as min. in accordance to notes specified in Doc.No. PDRP-4200-8320-SP-1001;
   - All nuts, bolts and washers shall be Hot Dip Spun Galvanized.
   - Erection Bolts : Min. 2 Nos M20 dia
   - Structural Connection Bots : Min 2 Nos M20dia

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- Stair Treads Connection: 2nos-M12; Ref. "Section 5; Stairs" of Doc. No. PDRP5200-8320-Sp-1001
- Handrail Connection: 2nos-M16; Ref. "Section 5 & 6; Stairs & Handrailing" of Doc. No. PDRP5200-8320-Sp-1001
- Nuts, Bolts, Washers shall be as per ;

Item Description	Material Code
High Strength bolt (HSB)	Grade 8.8 Confirm to IS: 1367 ; Tested as per IS: 1608 ; HSB Bolts shall be threaded in accordance with IS 4218.
Nuts	Shall confirm to IS: 1363 & IS: 1364
Washer	Plain washers :: Made of MS conforming to IS: 2016 ; IS 5369 Helical spring washers :: Conforming to IS: 6755 (for dynamic applications) Tapered washers :: Conforming to IS: 5372 and IS: 5374 (for channels and beams with tapered flanges)

- All connection bolts shall be tightened to the **"SNUG TIGHT"** condition.
- Mild steel bolts of Grade 4.6 or similar shall not be used.

5.11 **Anchor Bolts** (Ref. PDRP-8310-SP-0003)

LTHE has developed Anchor Bolt standard drawing in compliance to various tender requirement. Reference shall be made to LTHE project specific standard drawing Std. Dwg. No. IPEG-6389-8310-39-002-0003.

Following to be noted in this regards as the basis;

a) Anchor blot shall turned from MS rounds & confirm to IS: 1367 III, minimum grade 4.6 or equivalent in compliance to Cl.17 of PDRP-8310-SP-0003
b) Anchor Bolts shall be designed for 3mm corrosion allowance in compliance to Cl.9 of PDRP-8310-SP-0003
c) Allowable stresses in bolt shall be increased by 25% for short term loading condition i.e. under Wind & Seismic case of design
d) Threads shall be I.S.O. Metric Screw threads, coarse series, free fit, Class 7H/8g conforming to IS 4218
e) Bolts having dia > 64mm shall have constant 6mm pitch

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f) Nuts shall be tightened to a "Snug Fit" and then locked with lock nut where ever required

g) All anchor bolts shall have two Hexagonal Nuts (1no-Heavy Hex Nut & 1no- Lock Nut) as minimum unless specified otherwise in engineering drawing

h) Hexagonal Nuts & Lock nuts shall be property Class-4.6 conforming to IS: 1363 (Part-III) 7 IS: 1364 (Part-IV) respectively. Mechanical properties shall be as per IS: 1367.

i) Plain washer shall be Type-A conforming to IS 2016 ; IS 5569

j) Plate washer where ever required shall be of Grade E250 Grade-B conforming to IS 2062

k) Bolt Sleeve shall be fabricated from Carbon Steel Schedule 40 Pipe (Galvanized)

l) High Strength bolts; AQSTM Grade shall be used wherever required in view of design limitations with prior approval from IOCL as per Cl. 8.2 of PDRP-8310-SP-0003.

m) Minimum bolt spacing shall be in compliance to Cl.15 of PDRP-8310-SP-0003 ;
6 x bolt diameter: Vertical towers, columns and drums
4 x bolt diameter: Horizontal drums, exchangers and structural steelwork

n) Plinth edge distance shall be in compliance to Cl.13 of PDRP-8310-SP-0003

5.12 **Insert Plates** (PDRP-8310-45-0001 "Paving & U/G Services General Notes & Legend")

Insert Plates (embedded plates) shall be of Grade E250A conforming to IS: 2062-2011.

Insert plates shall be hot dip galvanized in accordance to applicable codes and standard. Damaged steel surface due to welding shall be painted with approved make galvanize paint after proper cleaning of welded surface.

5.13 **Grating**
(Cl.11 "Grating" Doc. No. PDRP-8320-SP-0016)
(PDRP4200-8320-SP-1001 "Standard Steel Work Notes and Details")

LTHE has developed project specific grating standard drawing in compliance to tender specification as listed here under. (Reference shall be made to LTHE project specific standard drawing Std. Dwg. No. IPEG-6389-8310-39-002-0003 & IPEG-6389-8310-39-002-0009)

Grating span shall not exceed 1200mm. Allowable deflection shall be L/200 or 6mm whichever is minimum.

Grating shall be Electro-forged hot dip galvanized MS grating in compliance to following;

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a) General Usage as Floor :

- Bearing & Edge Bar : 25x5mm Grade E250A in compliance to IS 2062 spaced @ 30mm c/c
- Cross Bar : 6mm (twisted) MS Square bar in compliance to IS 1786 (2008) ; IS 1732 (1989). Alternatively vendor may use IS 7887 (equivalent to SAE 1008) as base material for making twisted bar as per proprietary manufacturing & design process of vendor. Spacing of cross bars shall be 100mm c/c.
- Galvanization : Min. thickness of galvanization shall be 900 GSM/Sq.m (120 microns)
- Grating Nuts & Botls : shall be Electroplated GI confirming to ASTM a307 & ASTM A563
- Grating Clamps : material shall conforming IS 2062 Gr.A/b, to be hot dip galvanized with avg. 460 gm/Sq.m (65 microns)

b) Heavy Duty Grating Covers: Used as cover for drain & Catch basin shall have 35x6thk flats placed @ 25 c/c with ISA 40x40x5 edge angle.

c) Steel Stair Tread : 25x5mm Grade E250A in compliance to IS 2062 spaced @ 30mm c/c with 6mm (twisted) cross bars @ 100mm c/c spacing in compliance to IS 1786 (2008) ; IS 1732 (1989). Alternatively vendor may use IS 7887 (equivalent to SAE 1008) as base material for making twisted bar as per proprietary manufacturing & design process of vendor.

5.14 **Fire Proofing**

(Ref. Doc.No. PDRP-8440-SP-0007)
(Ref. Doc.No. PDRP-8310-SP-0011 "Concrete Fire Protection)

Fireproofing shall have 2 hurs fire rating. Steel structures shall be fireproofed upto 2m from HPP with Cast In-Situ dense concrete for vehicular protection, above which light weight vermiculite fire proofing shall be applied.

a) Cast In-Situ formed concrete :

- Shall be 50mm, M30 grade concrete as is IS456 with minimum cement content of 320 kg/cu.m
- Cement shall be OPC in accordance with IS 8112, IS 12269 or Sulphate resisting cement in accordance with IS 12330
- Nominal maximum size of aggregate shall be 10mm down in accordance with IS 383

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- Welded galvanized wire fabric with min. 2.5mm dia cross wires at 50mm centers.
- Tying wire shall be 1.6mm soft annealed galvanized wire
- Nelson T-stud or 20mm square nuts shall be welded on steel structure as per standard drawing

b) Vermiculite :

- Thickness of fireproofing shall be 34mm for profile & 29mm for solid fireproofing for 2hr fire rating for Fendolite mil (make : Cafco / Newkem) having density 775 kg/cu.m
- In case of Pyrocrete 241 (Carboline) thickness of 29mm shall be followed for profile as well solid fireproofing, having 881 kg/cu.m density
- Plastic coated 1mm dia galvanized wire mesh, 50mm x 50mm hexagonal mesh
- Tying wire shall be 1.6mm soft annealed galvanized wire
- Nelson T-stud or 20mm square nuts shall be welded on steel structure as per standard drawing

6. LOADS AND FORCES

6.1 General

The below mentioned basic loads shall be applied to all the structures based on applicability, irrespective of the material of construction. The design shall be checked against the various combinations. The following type of basic loads and forces shall be considered for design

Dead load (including self-weight)	"DL"
Live load	"LL"
Equipment load	
Equipment load for erection	"E(E)"
Equipment load for operating	"E(O)"
Equipment load for testing/flushing	"E(T)"
Piping load	
Piping load for erection	"P (E) "
Piping load for operating	"P(O)"
Piping for testing/flushing	"P(T)"
Impact load & Vibratory load	"VL"

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Wind load	"WL"
Seismic load	"SL"
Thermal Anchor force	"TA"
Thermal Friction force	"TF"
Tube bundle pulling force	"BP"
Earth pressure	"EP"
Process Vibration (Surge)	"S"

Detailed calculation of the application of all the above applicable loads shall suitably be covered in design calculation document of individual item.

6.2 Dead Loads (DL)

Dead Load shall be considered as the total weight of the materials forming a permanent part of the structure / unit. This includes the self-weight of RC / Steel structure, Fire proofing-insulation, Grating, Handrail, Stair, Ladders, and Equipment rigidly fastened to the structures, piping, cables etc which are not movable / removed during its service life.

Component of soil backfill weight over foundation slab shall be considered as foundation dead load.

Self-weight of the member is automatically calculated by STAAD.Pro according to the applied density. The unit weight of materials in general, should be in accordance with IS: 875 (Part-1). However, for following materials, minimum unit weights shall be as follows:

Reinforced Concrete	25 kN/m^3
Plain Concrete	24 kN/m^3
Structural Steel	78.5 kN/m^3
Backfill Soil	18 kN/m^3
Brick Walls (unplaster)	19 kN/m^3
Grating 25mm thk (if beams & joist are modeled in STAAD for actual design)	0.5 kN/m^2
6 mm Thk. Chequered plate	0.55 kN/m^2
Staircase (Steel)	1.4 kN/m^2

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Ladder	0.4 kN/m
Handrail	0.15 kN/m
False Ceiling	0.75 kN/m^2 (or as per vendor data)
Fireproofing Concrete - Dense Concrete - Vermiculite	 24 kN/m^3 8.81 kN/m^3

In addition to above, estimated actual load from electrical trays and instrumentation ducts shall be considered at the specified locations, together with walkways, if provided.

6.3 Live Loads (LL)

There shall be two component of Live Load i.e. leading Live Load & Accompanying Live Load.

a. Leading Live Load :

Referring to Table-4, of IS 800:2007, leading live load shall be considered to be the one causing the higher load effect in the analysis and design of the structure.

Accordingly, Equipment & Piping content weight shall considered as Leading Live Load and is being accounted as separate load case under Equipment Operating E(O) / Pipe Operating P(O) Load here under in this document.

b. Accompanying Live Load :

Referring to Table-4, of IS 800:2007, accompanying live load shall be considered to be the one causing the lower load effect.

Accordingly, the live load attributed due to weight of all movable loads i.e. personnel, tools, miscellaneous equipment for O&M purpose, temporarily stored material etc which is quite temporary and secondary in nature viz a viz the primary content weight of equipment & piping load for the process structure, is considered as accompanying load due to virtue of its nature of actual application.

Accordingly, is being defined as Accompanying Live Load for all analysis & design consideration of Civil, Structure & Building works.

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The live load shall be uniformly distributed over the horizontal projection of the specified areas.

Live loads shall be in compliance to IS 875: Part-II, however, the following minimum live loads should be considered in the design of structures to comply with tender requirement;

	Process Building / Technological Structure (Open/Enclosed type)	**Intensity of Live load (LL)**
1.	Operating	5.0 kN/m^2
	Maintenance	7.0 kN/m^2
	Pump / compressor house	7.5 kN/m^2
	Hand rail	0.36 kN/m
2.	**Service Platform**	
	Access Way	2.5 kN/m^2
	Vessel / Tower	2.5 kN/m^2
	Isolated Platform (for Valve Operation)	2.5 kN/m^2
	Pipe rack Walkways	2.5 kN/m^2
3.	**Substation / Control room**	
	Panel Floor / battery room	10.0 kN/m^2
	Control room / Other Areas	5.0 kN/m^2
	Partition	1.0 kN/m^2
	Concrete roofs with access for maintenance only	1.5 kN/m^2
	Roofs used for access to equipment	2.5 kN/m^2
	Sheeted roofs with access for maintenance only	0.75 kN/m^2
4.	Laboratory	4.0 kN/m^2
5.	Warehouses and open storage areas	7.5 kN/m^2
6.	**Staircase**	
	Process building / Technological Structure	5.0 kN/m^2
	Substation / Control room	3.0 kN/m^2
7.	**Walkway**	

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	Gantry Girder	3.0 kN/m^2
8.	False Ceiling	0.75 kN/m^2
9.	False Flooring	3.1 kN/m^2

Further to the above table, following shall also be considered as applicable.

1	Road / Bridges / Culverts	As per IRC Class AA loading / class 'A' loading or crane / vehicle loading (relevant for the plant) whichever is critical.
2	Underground Tanks/ Pits / Trenches	To be checked for Uplift assuming Ground - Water Table raised up to Finished Grade Level. Live load shall be accounted as per the pit configuration based on sand filling / operating platform etc as defined in the Plot Plan.

Live load on various types of roofs should be as per the requirements given in IS: 875 (Part-II).

- Ref. Cl. 7.3.2 of IS 1893 (Part-1) 2006; live load on roof shall be considered as zero for seismic mass calculation as well for design purpose.
  However, equipment load if any on the roof shall be added to the dead load of the building and shall not be reduced for earthquake force calculations.
- Ref. to Table-10 & Cl. 8.4.1 of IS1893 (Part 4) 2015; live load reduction of 25% & 50% shall be considered for live loads ranging from 3 KN/Sq.m & 5 KN/Sq.m as applicable for respective structures and its seismic load calculations & global design of structure with appropriate load combinations.
- However, grating supporting joist shall be designed locally for intended live load as specified in the table above.

6.4 **Wind Load (WL)**

Wind load on plant buildings / structures shall be calculated in accordance with IS 875: Part-3 (2015).

Typical Wind Pressure:

Wind pressures (Pd) are tabulated for this site as under. For intermediate heights linear interpolation of coefficient k2 as permitted by IS: 875 shall be used.

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Basic parameters for the site are as under.

Basic wind speed,	Vb = 55 m/s (Cl. No. 4.3.1 of PDRP-8310-SP-0001)
Risk coefficient,	k1 = 1.0 (table 1 of IS: 875 - Part 3)
Terrain height/Structure size factor,	k2 (as per table 2 of IS: 875 - Part 3, Category 2, in line with the tender documents)
Topography factor	k3 = 1.0 (Cl. No. 6.3.3 of IS: 875 - Part 3)
Importance factor for cyclonic region	k4 = 1.15 (Cl. No. 4.3.4 of PDRP-8310-SP-0001)
Design wind speed	Vz = Vb x k1 x k2 x k3 (Cl. No. 6.3 of IS: 875-Part 3) = 55 x 1 x k2 x 1.15 = 63.25 k2

(Ref. 3.1.1 of PDRP-8820-SP-0001, Basic wind speed 65 m/sec i.e. 55m/sec x 1.15)

Vz ~ 65 k2 m/sec is considered conservatively

Design wind pressure $Pd = 0.6 * Vz^2$

a. Open Frame Structure :

Wind Load on open-structures shall be calculated on the basis of areas of obstructions to the wind-flow with appropriate shielding effect. Wind force on structural elements is automatically calculated by STAAD.Pro based on frontal area normal to wind direction and defined wind pressure at various heights. The wind pressures defined in STAAD are arrived by multiplying the force coefficient and Design Wind Pressure Pd as calculated in table below;

Height z (m) above FGL	k2 for Terrain Category-2	Vz (m/s)	Pd (kPa)
< or equal to 10 m	1.00	65.0	2.53
15	1.05	68.25	2.79
20	1.07	69.55	2.90
30	1.12	72.80	3.17
50	1.17	76.05	3.47

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b. Pipe Rack Structure:

Transverse wind loading shall be calculated corresponding to the width of the pipe rack as per following table. This force shall be considered irrespective of height between two tiers.

Width of Pipe Rack	**Wind force at each tier level**
Up to 4.0 m	1.25 x p x s
Above 4.0m but up to 6.0m	1.50 x p x s
Above 6.0m but up to 10.0m	2.00 x p x s
Above 10.0 m	To be calculated at actual as per frontal area

Where;

Pd = 0.6 $(65 \times k2)^2$ Kg/m²; Design wind pressure as shown above

s = spacing of portals (m)

H = Projected height (m)

Above mentioned wind force formulae includes the wind forces on structure and pipe.

Wind load on Flare Header, elevated supports for supporting pipes, cable tray support structure etc. above pipe rack top tier elevation shall be evaluated as per their actual frontal area calculation and applicable wind pressure at that height.

Wind loads as calculated above shall be applied at the junction of column & longitudinal beam.

Wind force shall be calculated based on the following formula.

F = Projected area x Wind pressure x shape factor

6.5 Seismic Load (SL)

Seismic Loads on buildings / structures / foundations should be calculated in accordance with IS1893 (Part 1):2016 & IS1893 (Part 4):2015, Part 4-2015 as applicable for various plant facilities.

Plant Location : Paradip Orissa, India

Seismic Zone : III

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Zone Factor (Z) : 0.16

Design Spectrum : Annex-B of IS 1893 (Part 4)2015

(Note : Ref. Annexure-1 of this document, LTHE is considering conservative of the two i.e. Codal Spectra Vs. Spectra suggested by M/s FUGRO in 2008 report (widely used by all LSKT contractor in Paradip refinery), as the spectra value for design of plant facilities for LSTK-1 & LSTK-2 scope of work awarded to LTHE.

LTHE is further developing site specific spectra as per the tender scope of work, and shall validate the same with above assumption and necessary design updates as needed shall be carried out by LTHE)

Referring to Cl.10 of Doc. No. IEGP-6351-8110-RA-000-0001 Rev.T1 "RA Methodology & Assumptions", Conclusion & Recommendations as narrated in the tender document is extracted as under;

The study reveals that for most of the failure cases, hazard distances are limited to plant boundary and not affecting outside the complex. The detailed discussion on risk analysis of the identified failure cases has been given in Section 9.

According to above conclusion & recommendations, "failure/leakage can lead to directly or indirectly to extensive loss of life / property to population at large in the area adjacent to the plant complex" is not envisaged by contractor any of the LSTK-1 & LSTK-2 facilities.

Accordingly, in compliance to Table-6 & Table-3 of IS1893 (Part 4)2015 structure classification & its importance factor has been tabulated here under for analysis and design of various plant facilities under LSTK-1 & LSTK-2 scope of work by LTHE ;

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Item tag	Seismic categorization	Importance Factor, I
Process Plant Facilities :		
Pipe Rack With Or Without Air Fin Cooler	2	1.5
Technological / Engineering Structures	2	1.5
RC Elevated Table Top Structure For EO Reactor	2	1.5
Compressor House / Shelter / Sheds	2	1.5
Equipment Foundations : Process Columns, Exchanger, Vessel, Drums etc	2	1.5
Compressor, Blowers & Pumps	2	1.5
Package Foundations	2	1.5
Pipe Supports & Stanchions	2	1.5
Operating Stages & Platform around equipment for O&M purpose only	3	1.25
Rain Protection Shed For Pits	4	1.0
Offsite & Utility Facilities :		
Offsite Pipe Rack	3	1.25
Unit Rack	2	1.5
Unit / Offsite Rack Supporting Flare Line	2	1.5
Flare Trestle	2	1.5
Technological / Engineering Structures	2	1.5
Compressor House / Shelter / Sheds	2	1.5
Equipment Foundations : Process Columns, Exchanger, Vessel, Drums etc	2	1.5
Compressor, Blowers & Pumps	2	1.5
Package Foundations	2	1.5
Pipe Supports & Stanchions	2	1.5
Operating Stages & Platform around equipment for O&M purpose	3	1.25
Ethylene Storage Sphere : Foundations	1	2
EG / Hydrocarbon Storage Tank Foundations	2	1.5
Process Water Storage Tank Foundations	3	1.25
EG Gantry	2	1.5
Buildings : (Process Unit & Offsite)		
SubStation Building	2	1.5

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Satellite Rack Room (Blast Resistant / RC Frame Structure)	2	1.5
S&D Building	3	1.25
PP Laboratory Building	3	1.25
Sample Retention Room	3	1.25
Fire Water Booster Pump Shed / House		
Cooling Tower Basin, Structure & Foundation	3	1.25
Cooling Tower : Chemical House	3	1.25
TEG Barrelling Shed	3	1.25
Toilet Block	4	1.00
Parking Shed (Steel / RC)	4	1.00

Above is the list of major items in contractor scope of work for which basis shall be as defined above in compliance to tender specifications.

Other misc. Items not covered above, shall be considered in compliance to IS: 1893 (Part 4) : 2015, which shall be reflected in the respective design calculations.

Ref. to Note-1 of Table-3 IS 1893 (Part 4):2015, when structures are designed using site spectra corresponding to MCE, importance factor "I" shall be considered as Unity for all structure.

Damping - Damping ratio coefficients for different construction material shall be in line with the table 4 of IS: 1893 (Part 4 - 2015). The same has been reproduced as below:

Structure/Component	MCE	DBE
Steel structures	0.04	0.02
Reinforced concrete	0.07	0.05

For combined structures, damping ratio coefficient shall be determined based on well established procedures. If composite damping ratio is not evaluated, it shall be taken as that corresponding to material having lower damping.

Response Reduction Factor "R" -

Response Reduction Factor "R" shall be considered in compliance to Table-4 of IS 1893 (Part 4):2015.

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Further referring to Cl.8.6 of IS 1893 (Part 4): 2015, ratio of (I/R) is recommended as Unity for mechanical design of equipment and its components including anchor bolts as applicable. (Reference shall be made to Mechanical Design Basis)

However, as specified in Cl. 8.6 of IS 1893 (Part 4) : 2015 value of "R" for Equipment (Column, Vessels, Exchanger, Drums, Pumps, Compressor, Blower etc) supported on Steel Structure & RC Elevated Table Top structure shall be considered as per Table-4 of IS 1893 (Part 4):2015.

Response reduction factor "R" for RC Block Foundation / elevated pedestals (for Column, Vessels, Exchanger, Drums, Pumps, Compressor, Blower etc) shall be considered as "R = 2.5" on conservative side against value of R=3 as defined in Table 4 against Ordinary Shear wall with OMRF.

Seismic analysis and design of the structure shall be done by CQC method. Mass participation ratio shall be up to 90% as per IS 1893 (Part 4) 2015. Equipment-Structure interaction where ever required shall be done in compliance to IS 1893 (Part 4) : 2015.

6.6 Equipment Load

Equipment (Rotating, Static, Packages etc) loading data are considered as per the loading data received from the Mechanical department / Vendor's as applicable.

Equipment loads are mainly classified into following three cases;

Equipment load for Empty/Erection "E(E)":
This is the 'dead' weight of vessels, columns, etc., completely installed (including all fixtures, platforms, ladders, piping, internals, insulation and fireproofing etc as applicable) and ready for operation, however, without process fluid as designed. This load shall contribute to the dead load of the structure.

Equipment load for Operation "E(O)":
Operating loads shall be defined as the weight of equipment as defined in "E(E)" case + Process Liquids or solids with in equipment as intended during plant operation.

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Equipment load for Hydro Test "E(T)":
This includes weight of equipment E(E) + the weight of water filled inside equipment for hydro test purpose.

When hydrostatic pressure testing of equipment is required at site in-situ condition, the weight of equipment, completely filled with water, shall be incorporated in the design of the supporting structure.

In case of more than one number of large size equipment supported on single structure, and the hydro test case is the governing case, structure shall be designed for single largest equipment hydro test at a time. Appropriate design note shall be specified on the drawing.

Under hydro test condition the wind force shall be taken as 25% of normal wind loading.

6.7 Piping Load

Piping loads to be considered as actual, as per the Civil Design Information (loading data) provided by Piping Engineering group. Piping loads are classified in following three classes based on their conditions.

Piping Loads For Erection/Empty "P(E)":
This Includes self-weight of pipe with insulation, attached flanges & fittings, supports etc. Weight/loading of special valves (like equipment) shall be considered as per piping loading data information. This load shall contribute to the overall dead load of the structure.

Piping Loads For Operation "P(O)":
Operational load is defined as the weight of "Pipe + Liquid" contained within the pipe during operation. In case where the Civil Information drawings received from Piping Engineering mentions the total Operating load (i.e. Empty weight of pipe + Fluid weight), the same shall applied appropriately.

Piping Loads For Test "P(T)":
This includes weight of "Pipe + Weight of water filled inside pipe for hydro-test purpose" for the stipulated length of pipe system. In case where the Civil Information drawings received from Piping Engineering mentions the total Test load (i.e. Empty weight of pipe + Test Fluid weight), the same shall be applied appropriately.

Under hydro test condition the wind force shall be taken as 25% of normal wind loading.

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The above loads shall be considered in line with the inputs provided by Piping Engineering Department. Test loads should not be applied to more than one tier simultaneously.

Loading on Longitudinal Beams

Longitudinal beams connecting portal columns shall be sufficiently strong to sustain 25% of the load on the transverse beams. This total load shall be assumed as two equal concentrated loads acting at 1/3rd span. Other longitudinal axial forces coming on it from the design of the supporting system shall also be simultaneously taken into account in the design of the longitudinal beam. Friction and Anchor Forces, if specifically given by the piping specialist shall also be catered for in the design. Loads from monorails, when supported from these beams, shall also be considered to be acting simultaneously along with all other loads mentioned above.

6.8 Friction load (TF)

Friction load occurs as a result of friction between equipment/pipe and supports during thermal expansion. Friction force is calculated as operating load multiplied by friction coefficient between the contact surfaces. Friction coefficient (μ) for various types of contact surfaces are given below:

Teflon to Teflon	: 0.10
Teflon to steel	: 0.10
Steel to steel	: 0.30
Steel to concrete	: 0.45
Concrete to soil	: 0.40
Concrete to concrete	: 0.50

Following two instances shall be considered in design.

a) For Piping

Where the pipes are of similar diameter and service conditions, the friction force at each tier on every portal both in longitudinal and transverse directions shall be considered as follows.

No of pipes on one tier	:	Frictional force value
1 to 3 pipes	:	30 % of the design vertical load
More than 3 pipes	:	10 % of the design vertical load

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Longitudinal friction force (acting on the transverse beams) shall be considered as uniformly distributed load over the entire span of the beam at each tier and the transverse friction force (acting on the longitudinal beams) shall be considered as a concentrated load at each tier level.

Friction force on T-supports and trestles shall be taken as 30% and 10% in longitudinal and transverse directions respectively of the vertical loading. Both longitudinal and transverse friction forces shall be considered to be acting simultaneously.

For two-phase fluid flow/transfer lines frictional force shall be minimum 50% of the weight of pipe including contents & insulation, acting simultaneously in transverse & longitudinal direction.

Pipe Anchor & Guide loads (excluding their friction component) as per piping stress analysis report Referred in Cl. 5.9 of this document shall be combined with wind or seismic loads for all design cases including foundations as per Cl.2.8 of IEGP-6351-8310-SP-000-8006 Rev.T1 .

Estimated Frictional forces are temporary in nature till anchor point start restricting piping movement, and hence shall not be combined with short term loads viz. wind or seismic for Design of Piles / Foundation & Steel Structure. However, Steel & RC super structure (excluding foundations) shall be designed for "Thermal Friction + Thermal Anchor-Guide" forces under normal Operating condition.

b) For Equipment

Horizontal forces due to thermal expansion of horizontal vessels/ Exchangers shall be relieved by using slotted holes and slide plates and remaining force derived from product of the sliding saddle gravity load and co-efficient of friction (0.3 to steel or PTFE), which shall be applied to each support.

Horizontal load due to thermal expansion of heat exchangers and horizontal vessels shall be computed on top of the saddle supports under Equipment operating E(O) case as follows :

$H_{tf} = W_{os} * \mu$ where,

W_{os} = Fraction of operating weight of Equipment acting on sliding saddle. (This component shall be calculated based on the actual CG of the equipment)

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μ = Coefficient of friction

H_{te} = (0.5 x α x ΔT x L) x (3 x E x I / H_2^3) where,

α = Coefficient of linear expansion

ΔT = Temperature difference (Ambient and operating temperature on shell side.)

L = Saddle to Saddle distance

E = Young's modulus of Elasticity (Concrete)

= 5000 x sqrt (fck)

= 5000xsqrt (30) = 2.74x10^4 N/mm^2

I = Moment of Inertia of supporting RC pedestal

H2 = Height of Pedestal above top of footing.

Thermal force considered for foundation / pedestal design shall be minimum of thermal friction force Htf & Thermal expansion force Hte.

TF = Thermal force = Min (Hte, Htf)

6.9 Thermal Load - Anchor and Guide force (TA)

Thermal load shall be defined as those forces caused by a change in temperature. Such forces shall include those caused by expansion or contraction of piping / equipment supported over a structure.

The Anchor or Guide Forces in longitudinal and transverse directions shall be as per design inputs provided by Piping Engineering Group.

6.10 Bundle Pull (BP)

During maintenance of equipment bundle-pull loads are applicable. Bundle pull for different types of Exchangers should be taken as under:

- Fixed type – Nil
- Kettle type - 0.30 x Bundle weight.
- All other types - 0.86 x Bundle weight or 30N/mm of diameter whichever is greater.

Total Bundle Pull should be considered on fixed pedestal alone. Beams/frames under each saddle shall be equal in size, with equal settlements / deflections.

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In case of stacked exchangers, bundle pull resulted from the upper shell shall be considered for design. For a Technological structure, supporting more than one exchangers, only one bundle pull force causing maximum effect on overall structure shall be considered in the design of structures. However, for individual supporting systems, relevant bundle pulling force shall be considered for design.

Bundle pulling force is not applicable when bundles are pulled by means of a mechanical device, which acts on the principle of equilibrium of forces. In designing a structural member subject to several maintenance loads, consideration shall be limited to one crane or trolley load acting simultaneously with corresponding one tube bundle pulling load.

6.11 **Handling device load / Impact Load (HL) (For cranes, monorails etc.)**

Handling device loads shall mean the loads of Cranes, hoists and lifts including the lifted weight in the normal operation. Impact Load shall be defined as an equivalent static force caused by a moving object. The minimum specified vertical load due to equipment, machinery, crane or other objects that may produce impact or vibration is the total weight of the equipment or machinery plus its maximum lifting / carrying capacity, or the appropriate live load, multiplied by an impact factor.

Handling device loads shall be increased with the following percentages to include adequate allowance for the following impact conditions.

Vertical impact loads:
Frames supporting lifts and hoists: 100%
Foundations supporting lifts and hoisting apparatus: 40%
Vertical Loads for electric overhead cranes: 25%
Hand operated cranes: 10%

Horizontal impact loads:

(a) Overhead Cranes
Horizontal Loads transverse to rails: 20% of the maximum wheel loads of the crane applied at the top of rail.
Horizontal Loads along the rails: 10% of all static wheel loads

(b) Monorails
Transverse Load: 20% of the lifted load.

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Longitudinal Load: 10% of the lifted load hoist and trolley.

(c) Davit

Davits shall be designed for heaviest item to be lifted plus 25% for impact but not less than a total of 450 Kg.

Lateral force shall be assumed as 20% of the lifted weight.

6.12 Soil Pressure (SP)

Soil Pressure shall be taken into account for design of underground facilities such as foundation, retaining walls, trenches, drains, pits etc.

In the design of basement walls or similar structures below grade, provision shall be made for resisting lateral pressure of the adjacent soil. If a portion or whole of the adjacent soil is below ground water surface, pressure computations shall be based on the submerged weight of the soil (due to buoyancy) plus full hydrostatic pressure. Due allowance shall also be made for possible surcharges from fixed or moving loads subject to a minimum of 10 kN/m^2 as applicable.

Active earth pressure and Passive earth pressure coefficients shall be calculated as per the angle of internal friction of soil. In case of basement having RC tie -slab or pits with RCC roof, design shall consider soil pressure at rest.

Coefficient of earth pressure shall be in line with clause 10.0 of tender Geotechnical Investigation Report Doc. No. 213/08/R-1. Same shall further be updated as per final soil report.

Coefficient of friction between soil and concrete shall be considered as 0.4 in accordance with Cl. 5.12 of PDRP-8310-SP-0007 Rev.F1. Same shall further be updated as per final soil report.

Soil-Structure interaction shall be simulated in structural analysis wherever it is critical for either supporting structure or equipment.

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6.13 **Vibration Load (VL)**

Vibration loads shall mean the vibration forces caused by heavy vibrating equipment or machinery and dynamic forces caused by fluids in the normal operation.

Frames, structures and foundations for machinery or equipment causing vibration shall be designed to limit vibrations to an acceptable level.

It shall be designed such that whether they are independent or part of the building, it shall not only safely carry the loads for such items but also prevent resonance. Natural frequencies of frames, structures and foundations must differ by more than 20% from that of the machinery under operating conditions.

While carrying out dynamic analysis of foundations/structures supporting dynamically loaded equipment, the loads indicated by vendor in his documents shall be used. However, in the absence of such vendor data, the following loads shall be assumed as the equivalent static loads when the dynamic analysis is not carried out.

a) Rotating (Centrifugal) type

i) Vertical direction: 0.5 times the weight of equipment or machinery

ii) Horizontal direction

- Along direction of Rotation (Perpendicular to center line of shaft): 0.25 times the weight of equipment or machinery
- Along direction of shaft axis: 0.1 times the weight of equipment or machinery

b) Reciprocating Type

i) Vertical direction: 0.5 times the weight of equipment or machinery

ii) Horizontal direction

- Along direction of Reciprocating: 0.25 times the weight of equipment or motion machinery
- Along direction perpendicular to reciprocating motion: 0.1 times the weight of equipment or machinery

These horizontal loads shall be considered to be acting at shaft center line. In case elevation of shaft center line is not available, it shall be considered to be acting at EL (top of foundation + 500mm).

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6.14 **Contingency Load (CL)**

This load has been specifically considered for the design of Piperack Structure. This loads shall be considered in line with the details mentioned in clause 5.7.

6.15 **Blast Load (BL)**

EG-ERU Unit SRR-822 shall be designed as blast resistant / resilient building in compliance to Cl.5.0 of Doc. No. PDRP-8310-SP-0004 "Design Of Blast Resistant Building For High Hazardous Plant"

Satellite Rack Room (SRR) other than SRR-822 in LSTK-1 & LSTK-2 scope of work by LTHE shall be designed as Brick in-filled RC frame structure in compliance to IS 456 : 2005, IS 1893, IS 13920:2016.

Structures subjected to blast forces generated due to accidental blasts from hydrocarbon ignitions shall be designed to withstand a minimum pressure of 3psi or unless noted otherwise in the QRA report.

6.16 **Other Loads**

Apart from the above specified loads, any other loads or possible overloading due to particular requirements of erection, maintenance and construction, shall also be considered in the design, wherever applicable.

Design of all structures should also consider any other relevant and consequential load/stress imparted to the structure.

7. COMBINATION OF LOADS

All buildings / structures should be designed to resist the worst combination resulted from the above loads. It should be recognized in load combinations that the simultaneous occurrence of maximum values of wind & earthquake load is not likely.

Limit State Method as per IS: 456-2000 shall be followed for the design of RCC structure.

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All Structural Steel work shall be designed in accordance with the IS: 800-2007 Limit state method. .

Liquid retaining / storage RCC structure shall be designed by Limit State method with appropriate crack check as per IS 456:2000 / uncracked section as per IS: 3370.

Appropriate Live load reduction shall be considered in line with the IS: 1893 for the load combinations involving seismic loads.

These loading combinations are indicative only. For more detailed loading combinations individual STAAD Design file shall be referred.

7.1 **Loading Combinations for Blast resistant SRR Building:**

a) Load combinations for Deflection of superstructure, Soil Bearing Pressure / Pile Capacity Check and Stability of Foundation check ;

Load Condition	**Load Combination**
Normal	DL + LL
Wind	DL ± WL
	DL + 0.8 [LL ± WL]
Seismic	DL ± SL
	DL + 0.8 [0.5LL + SL]
Blast	DL + LL ± BL

b) Load combinations for RCC member design ;

Load Condition	**Load Combination**
Normal	1.5 [DL + LL]
Wind	1.5 [DL ± WL]
	1.2 [DL + LL ± WL]
	0.9 DL ± 1.5 WL
Seismic	1.5 [DL ± SL]
	1.2 [DL + 0.5 LL ± SL]
	0.9 DL ± 1.5 SL
Blast	1.0 [DL+ LL ± BL]

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7.2 Loading Combinations for Buildings : (Brick In-filled RC Frame Structure)

c) Load combinations for Deflection of superstructure, Soil Bearing Pressure / Pile Capacity Check and Stability of Foundation check ;

Load Condition	Load Combination
Normal	DL + LL
Wind	DL ± WL
	DL + 0.8 [LL ± WL]
Seismic	DL ± SL
	DL + 0.8 [0.5LL + SL]

d) Load combinations for RCC member design ;

Load Condition	Load Combination
Normal	1.5 [DL + LL]
Wind	1.5 [DL ± WL]
	1.2 [DL + LL ± WL]
	0.9 DL ± 1.5 WL
Seismic	1.5 [DL ± SL]
	1.2 [DL + 0.5 LL ± SL]
	0.9 DL ± 1.5 SL

7.3 For Vertical Column / Vessel / Horizontal vessel / Exchanger Foundations

a) Load Combinations for Soil Bearing Pressure / Pile Capacity Check and Stability of Foundation ;

Load Condition	Load Combination
Empty	DL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
Operating	DL + P(O) + E(O)
	DL + P(O) + E(O) ± WL
	0.9 [DL + P(O) + E(O)] ± WL
	DL + P(O) + E(O) ± SL
	0.9 [DL + P(O) + E(O)] ± SL
Test	DL + P(T) + E(T)

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	DL + P(T) + E(T) ± 0.25WL
Maintenance*	DL + P(E) + E(E) + BP

* Applicable for Exchangers only

b) Load Combinations for RC Pedestal / Foundation / Pile cap Design ;

Load Condition	Load Combination
Empty	1.5 [DL + P(E) + E(E)]
	1.5 [DL + P(E) + E(E) ± WL]
	0.9 [DL + P(E) + E(E)] ± 1.5WL
Operating	1.5 [DL + P(O) + E(O)]
	1.2 [DL + P(O) + E(O) ± WL]
	0.9 [DL + P(O) + E(O)] ± 1.5WL
	1.2 [DL + P(O) + E(O) ± SL]
	0.9 [DL + P(O) + E(O)] ± 1.5SL
Test	1.5 [DL + P(T) + E(T)]
	1.5 [DL + P(T) + E(T) ± 0.25WL]
Maintenance*	1.5 [DL + P(E) + E(E) + BP]

* Applicable for Exchangers only

7.4 For Structural Steel Pipe Rack / Technological Structures

a) Load Combinations for Stability Check of foundation / Soil Bearing Pressure for Foundation / Pile Capacity check ;

Load Condition	Load Combination
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 LL + P(E) + E(E) ± 0.8 WL
	DL + P(E) + E(E) ± SL
	DL + 0.8 [0.5LL] + P(E) + E(E) ± 0.8 SL
Operating	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± WL
	DL + 0.8 LL + P(O) + E(O) ± T(A) ± 0.8 WL
	0.9 [DL + P(O) + E(O) ± T(A)] ± WL
	DL + P(O) + E(O) ± T(A) ± SL
	DL + 0.8 [0.5LL] + P(O) + E(O) ± T(A) + 0.8 SL

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	0.9 [DL + P(O) + E(O) ± T(A)] ± SL
Test	DL + LL + P(T) + E(T)
	DL + LL + P(T) + E(T) ± 0.25WL
Maintenance	DL + LL + P(E) + E(E) + CL + BP

b) Load Combinations for RC Pedestal/Foundation/Pile cap Design ;

Empty	1.5 [DL + LL + P(E) + E(E)]
	1.5 [DL + P(E) + E(E)] ± 1.5 WL
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 [DL + LL + P(E) + E(E)] ± 1.2 WL
	1.5 [DL + P(E) + E(E)] ± 1.5 SL
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
	1.2 [DL + 0.5LL + P(E) + E(E)] ± 1.2 SL
Operating	1.5 [DL + LL + P(O) + E(O) ± T(A) ± T(F)]
	1.5 [DL + P(O) + E(O) ± T(A) ± WL]
	0.9 [DL + P(O) + E(O) ± T(A) ± 1.5WL
	1.2 [DL + LL + P(O) + E(O) ± T(A) ± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± SL]
	1.2 [DL + 0.5LL + P(O) + E(O) ± T(A) ± SL]
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + P(T) + E(T) ± 0.25WL]
	1.2 [DL + LL + P(T) + E(T) ± 0.25WL]
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL + BP]

c) Load Combinations for Beam Deflection Check and Column Sway Check

Load Condition	**Load Combination**
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 LL + P(E) + E(E) ± 0.8 WL
	DL + P(E) + E(E) ± SL
	DL + 0.8 [0.5LL] + P(E) + E(E) ± 0.8 SL
Operating	DL + P(O) + E(O) ± T(A) ± T(F)
	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± T(F) ± WL
	DL + 0.8 LL + P(O) + E(O) ± T(A) ± T(F) ± 0.8 WL

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	DL + P(O) + E(O) ± T(A) ± T(F) ± SL
	DL + 0.8 [0.5LL] + P(O) + E(O) ± T(A) ± T(F) ± 0.8 SL
Test	DL + LL + P(T) + E(T)
	DL + LL + P(T) + E(T) ± 0.25WL
Maintenance	DL + LL + P(E) + E(E) + CL + BP

d) Load Combinations for Steel Member Design

Empty	1.5 [DL + P(E) + E(E)] +1.05 LL
	1.2 [DL + P(E) + E(E)]
	0.9DL + 1.2 [P(E) + E(E)]
	1.2 DL + 1.05LL + 1.2 [P(E) + E(E)] ± 0.6WL
	1.2 DL + 0.53LL + 1.2 [P(E) + E(E))± WL]
	1.5 [DL + P(E) + E(E) ± WL]
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 DL + 1.05 [0.5LL] + 1.2 [P(E) + E(E)] ± 0.6SL
	1.2 DL + 0.53 [0.5LL] + 1.2 [P(E) + E(E) ± SL]
	1.5 [DL + P(E) + E(E) ± SL]
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
Operating	1.5 DL + 1.05LL + 1.5 [P(O) + E(O) ± T(F) ± T(A)]
	1.2 DL + 1.05LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6WL
	1.2 DL + 0.53LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± WL]
	0.9 [DL + P(O) + E(O) ± T(A) ± T(F)] ± 1.5 WL
	1.2 DL + 1.05(0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6SL
	1.2 DL + 0.53(0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)± SL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± SL]
	0.9 [DL + P(O) + E(O) ± T(A) ± T(F)] ± 1.5 SL
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + LL + P(T) + E(T) ± 0.25 WL]
	0.9 [DL + P(T) + E(T)] ± 1.5 (0.25 WL)
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL + BP]

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7.5 For Reinforced Concrete Table Top Structures

a) Load Combinations for Stability Check of foundation / Soil Bearing Pressure for Foundation / Pile Capacity check

Load Condition	Load Combination
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 LL + P(E) + E(E) ± 0.8 WL
	DL + P(E) + E(E) ± SL
	DL + 0.8 [0.5LL] + P(E) + E(E) ± 0.8 SL
Operating	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± WL
	DL + 0.8 LL + P(O) + E(O) ± T(A) ± 0.8 WL
	0.9 [DL + P(O) + E(O) ± T(A)] ± WL
	DL + P(O) + E(O) ± T(A) ± SL
	DL + 0.8 [0.5LL] + P(O) + E(O) ± T(A) + 0.8 SL
	0.9 [DL + P(O) + E(O) ± T(A)] ± SL
Test	DL + LL + P(T) + E(T)
	DL + LL + P(T) + E(T) ± 0.25WL
Maintenance	DL + LL + P(E) + E(E) + CL + BP

b) Load Combinations for RC Pedestal/Foundation/Pile cap Design:

Empty	1.5 [DL + LL + P(E) + E(E)]
	1.5 [DL + P(E) + E(E)] ± 1.5 WL
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 [DL + LL + P(E) + E(E)] ± 1.2 WL
	1.5 [DL + P(E) + E(E)] ± 1.5 SL
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
	1.2 [DL + 0.5LL + P(E) + E(E)] ± 1.2 SL
Operating	1.5 [DL + LL + P(O) + E(O) ± T(A) ± T(F)]
	1.5 [DL + P(O) + E(O) ± T(A) ± WL]
	0.9 [DL + P(O) + E(O) ± T(A) ± 1.5WL
	1.2 [DL + LL + P(O) + E(O) ± T(A) ± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± SL]
	1.2 [DL + 0.5LL + P(O) + E(O) ± T(A) ± SL]

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Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + P(T) + E(T) ± 0.25WL]
	1.2 [DL + LL + P(T) + E(T) ± 0.25WL]
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL + BP]

c) Load Combinations for Beam Deflection Check and Column Sway Check

Load Condition	Load Combination
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 LL + P(E) + E(E) ± 0.8 WL
	DL + P(E) + E(E) ± SL
	DL + 0.8 [0.5LL] + P(E) + E(E) ± 0.8 SL
Operating	DL + P(O) + E(O) ± T(A) ± T(F)
	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± T(F) ± WL
	DL + 0.8 LL + P(O) + E(O) ± T(A) ± T(F) ± 0.8 WL
	DL + P(O) + E(O) ± T(A) ± T(F) ± SL
	DL + 0.8 [0.5LL] + P(O) + E(O) ± T(A) ± T(F) ± 0.8 SL
Test	DL + LL + P(T) + E(T)
	DL + LL + P(T) + E(T) ± 0.25WL
Maintenance	DL + LL + P(E) + E(E) + CL + BP

d) Load Combinations for Steel Member Design

Empty	1.5 [DL + P(E) + E(E)] +1.05 LL
	1.2 [DL + P(E) + E(E)]
	0.9DL + 1.2 [P(E) + E(E)]
	1.2 DL + 1.05LL + 1.2 [P(E) + E(E)] ± 0.6WL
	1.2 DL + 0.53LL + 1.2 [P(E) + E(E))± WL]
	1.5 [DL + P(E) + E(E) ± WL]
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 DL + 1.05 [0.5LL] + 1.2 [P(E) + E(E)] ± 0.6SL
	1.2 DL + 0.53 [0.5LL] + 1.2 [P(E) + E(E)± SL]
	1.5 [DL + P(E) + E(E) ± SL]
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
Operating	1.5 DL + 1.05LL + 1.5 [P(O) + E(O) ± T(F) ± T(A)]

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	1.2 DL + 1.05LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6WL
	1.2 DL + 0.53LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± WL]
	0.9 [DL + P(O) + E(O) ± T(A) ± T(F)] ± 1.5 WL
	1.2 DL + 1.05(0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6SL
	1.2 DL + 0.53(0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)± SL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± SL]
	0.9 [DL + P(O) + E(O) ± T(A) ± T(F)] ± 1.5 SL
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + LL + P(T) + E(T) ± 0.25 WL]
	0.9 [DL + P(T) + E(T)] ± 1.5 (0.25 WL)
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL + BP]

e) Load Combinations for RC superstructure Design:

Empty	1.5 [DL + LL + P(E) + E(E)]
	1.5 [DL + P(E) + E(E)] ± 1.5 WL
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 [DL + LL + P(E) + E(E)] ± 1.2 WL
	1.5 [DL + P(E) + E(E)] ± 1.5 SL
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
	1.2 [DL + 0.5LL + P(E) + E(E)] ± 1.2 SL
Operating	1.5 [DL + LL + P(O) + E(O) ± T(A) ± T(F)]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± WL]
	0.9 [DL + P(O) + E(O) ± T(A) ± T(F) ± 1.5WL]
	1.2 [DL + LL + P(O) + E(O) ± T(A) ± T(F) ± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± SL]
	1.2 [DL + 0.5LL + P(O) + E(O) ± T(A) ± T(F) ± SL]
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + P(T) + E(T) ± 0.25WL]
	1.2 [DL + LL + P(T) + E(T) ± 0.25WL]
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL + BP]

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7.6 For Compressor Shelter / Sheds

a) Load Combinations for Stability Check of foundation / Soil Bearing Pressure for Foundation / Pile Capacity check

Load Condition	Load Combination
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 LL + P(E) + E(E) ± 0.8 WL
	DL + P(E) + E(E) ± SL
	DL + 0.8 [0.5 LL] + P(E) + E(E) ± 0.8 SL
Operating	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± WL
	DL + 0.8 LL + P(O) + E(O) ± T(A) ± 0.8 WL
	0.9 [DL + P(O) + E(O) ± T(A)] ± WL
	DL + P(O) + E(O) ± T(A) ± SL
	DL + 0.8 [0.5 LL] + P(O) + E(O) ± T(A) ± 0.8 SL
	0.9 [DL + P(O) + E(O) ± T(A)] ± SL
Test	DL + LL + P(T) + E(T)
	DL + LL + P(T) + E(T) ± 0.25 WL
Maintenance	DL + LL + P(E) + E(E) + CL
	DL + 0.5 LL + P(E) + E(E) + CL ± 0.5 WL
	DL + 0.5 LL + P(E) + E(E) + 0.5 CL ± WL
	DL + LL + P(E) + E(E) + CL ± 0.5 SL
Note : For stability check of Foundation, Thermal Friction T(F) shall be neglected for Operating condition with wind and seismic	

b) Load Combinations for RC Pedestal/Foundation/Pile cap Design

Empty	1.5 [DL + LL + P(E) + E(E)]
	1.5 [DL + P(E) + E(E) ± WL]
	1.2 [DL + LL + P(E) + E(E) ± WL]
	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.5 [DL + P(E) + E(E) ± SL]
	1.2 [DL + 0.5 LL + P(E) + E(E) ± SL]
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
Operating	1.5 [DL + LL + P(O) + E(O) ± T(A) ± T(F)]
	1.5 [DL + P(O) + E(O) ± T(A) ± WL]

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	0.9 [DL + P(O) + E(O) ± T(A) ± 1.5WL
	1.2 [DL + LL + P(O) + E(O) ± T(A) ± WL]
	1.5 [DL + P(O) + E(O) ± T(A) ± SL]
	1.2 [DL + 0.5LL + P(O) + E(O) ± T(A) ± SL]
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + LL + P(T) + E(T) ± 0.25WL]
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL]
	1.2 [DL + LL + P(E) + E(E) + CL ± 0.5 WL]
	1.5 [DL + 0.5 LL + P(E) + E(E) + 0.5 CL ± WL]
	1.2 [DL + LL + P(E) + E(E) + CL ± 0.5 SL]

c) Load Combinations for Beam Deflection Check and Column Sway Check

Load Condition	**Load Combination**
Empty	DL + LL + P(E) + E(E)
	DL + P(E) + E(E) ± WL
	DL + 0.8 [LL] + P(E) + E(O) ± 0.8 WL]
	DL + P(E) + E(E) + SL
	DL + 0.8 [0.5 LL] + P(E) + E(O) ± 0.8 WL]
Operating	DL + LL + P(O) + E(O) ± T(A) ± T(F)
	DL + P(O) + E(O) ± T(A) ± T(F)± WL
	DL + 0.8 [LL] + P(O) + E(O) ± T(A) ± T(F)± 0.8 WL
	DL + P(O) + E(O) ± T(A) ± T(F)± SL
	DL + 0.8 [0.5 LL] + P(O) + E(O) ± T(A) ± T(F)± 0.8 SL
Test	DL + LL + P(T) + E(T)
	DL + P(T) + E(T) ± 0.25WL
Maintenance	DL + LL + P(E) + E(E) + CL
	DL + 0.5 LL + P(E) + E(E) + CL + 0.5 WL
	DL + 0.5 LL + P(E) + E(E) + 0.5 CL + WL
	DL + LL + P(E) + E(E) + CL + 0.5 SL

d) Load Combinations for Steel Member Design

Empty	1.2 [DL + P(E) + E(E)]
	0.9DL + 1.2 [P(E) + E(E)]
	1.2 DL + 1.05 LL + 1.2 [P(E) + E(E)] ± 0.6 WL
	1.2 DL + 0.53 LL + 1.2 [P(E) + E(E) ± WL]
	1.5 [DL + P(E) + E(E) ± WL]

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	0.9 [DL + P(E) + E(E)] ± 1.5 WL
	1.2 DL + 1.05 [0.5 LL] + 1.2 [P(E) + E(E)] ± 0.6 SL
	1.2 DL + 0.53 [0.5 LL] + 1.2 [P(E) + E(E) ± SL]
	1.5 [DL + P(E) + E(E) ± SL]
	0.9 [DL + P(E) + E(E)] ± 1.5 SL
Operating	1.5[DL + 1.05 LL + 1.5 [P(O) + E(O) ± T(F) ± T(A)]
	1.2 DL + 1.05 LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6WL
	1.2 DL + 0.53 LL + 1.2 [P(O) + E(O) ± T(A) ± T(F)± WL]
	1.5[DL + 1.05 LL + P(O) + E(O) ± T(A) ± T(F) ± WL]
	0.9 [DL + P(E) + E(E) + P(O) + E(O) ± T(A) ± T(F)] ± 1.5 WL
	1.2 DL + 1.05 (0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)] ± 0.6SL
	1.2 DL + 0.53 (0.5LL) + 1.2 [P(O) + E(O) ± T(A) ± T(F)± SL]
	1.5 [DL + P(O) + E(O) ± T(A) ± T(F) ± SL]
	0.9 [DL + P(E) + E(E) + P(O) + E(O) ± T(A) ± T(F)] ± 1.5SL
Test	1.5 [DL + LL + P(T) + E(T)]
	1.5 [DL + LL + P(T) + E(T) ± 0.25 WL
	0.9 [DL + P(E) + E(E) + P(T) + E(T)] ± 1.5 (0.25 WL)
Maintenance	1.5 [DL + LL + P(E) + E(E) + CL]
	1.2 [DL + LL + P(E) + E(E) + CL ± 0.5 WL]
	1.5 [DL + 0.5 LL + P(E) + E(E) + 0.5 CL ± WL]
	1.2 [DL + LL + P(E) + E(E) + CL ± 0.5 SL]

8. DETAIL DESIGN REQUIREMENT FOR FOUNDATION DESIGN

Reference shall be made to IPEG-6389-8310-PH-001-0002 "Design Basis For Foundation" for following;

- Type of foundation system adopted for various plan and offsite facilities in scope of LSTK-1 & LSTK-2 package
- Basis of foundation analysis & Design
- Recommendation for Safe Bearing Capacity (Shallow Foundation)
- Recommendation for Pile Capacity
- Detail analysis of Liquefaction and its mitigation plan
- Recommendation for Ground Improvement (for enhancement of Safe Bearing Capacity)

Reference to above, following to be followed as design guidelines for shallow / pile foundation;

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8.1 Shallow Foundation

Shallow foundation system shall be adopted for; medium to lightly loaded structure like Offsite Racks, pipe Tracks, Sleepers, Pipe Support & Stanchions, Pump-Blower-skid base package, Pits, Operating Platforms – Cross Overs, Single Storied RC/Steel Buildings & sheds, Analyzer Shelter etc. which can be easily founded on the Net Safe Bearing Capacity ranging from 5 T/Sq.m ~ 15 T/Sq.m for 25mm settlement on a virgin soil with appropriate compaction & PCC layer.

The allowable net bearing capacity for shallow foundations used in the design shall be minimum of safe bearing capacity based on shear-strength of soil and safe bearing capacity based on the settlement criteria as defined here under this section.

Basic design parameters for the shallow foundations shall be as follows:

a) Type of Foundation : Open / Shallow / Block / Isolated-combined Footing & Rafts

b) Depth of Foundation : 1.0m below HPP as min. depth of foundation
(Unless noted otherwise in engineering drawing)

c) Ground Water Table : Shall be considered at Finished Grade Level

d) S.B.C. considered : IPEG-6389-8310-PH-001-0002 "Design Basis For Foundation"

e) Allowable Increase : Net Bearing Capacity shall be increased by ;

- Wind Case :

  25% increase in SBC is allowed for all types of foundations i.e Shallow Isolated, Raft, Combined Raft, and Pile etc

- Seismic Case :

  25% increase in SBC is allowed for Shallow Isolated foundation

  50% increase in SBC is allowed for Isolated/combined Rafts

  50% increase in Pile Capacity for pile resting in firm strata having "N>30" as per IS 1893 (Part 4) 2015

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f) Settlement Criteria :

Sr. No.	Location	Settlement Criteria
1	Isolated Foundation : For all general usage	25 mm
2	Isolated Footing & Raft Foundation : Buildings (Note : If there are large isolated shallow foundation in building which can be classified as Isolated Raft foundation, for uniform settlement between normal footing and raft footing, 40mm allowable limit of settlement shall be considered for uniformity)	40 mm
3	Large Combined Raft Foundations	40 mm
4	Tank (Cl. 16 of PDRP-8310-SP-0008) - Total Permissible Settlement - Differential Settlement - Tilt Of Tank - Tank Floor Settlement along Radial Line from periphery to the tank center	 50mm 0.0015 L 1 : 500 1 : 300

Note: "L" denotes the length of deflected part of the Ring Beam Or Strip

g) Stability Against Overturning :

Ref. Cl. 5.3.2 of PDRP-8310-SP-0007 "Design Of Concrete Foundations, Structures & Paving"

Following shall be considered as general guidelines for Factor Safety (F.S.) against overturning i.e. Minimum Stability Ratio against overturning which is, "MR/MO", where;

MR = Resisting moment and MO = Overturning moment.

Load Combination description	Min Factor Of Safety Against Overturning
Long Term : Operating Load without Wind/Seismic Loads	2.00
Short Term : Operating Load + Wind/Seismic Loads	1.50

h) Stability against Sliding

The resistance to sliding shall be calculated from the following:-

i. Foundation / Soil friction (Friction coefficient between concrete to soil = 0.40)
ii. Passive soil resistance to the side of the foundation shall be ignored for small isolated shallow foundation.

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However, for large isolated / combined rafts, offering considerable soil resistant which can be considered confidently, can be considered in to account.

Passive soil resistance of soil bearing against keys.

Referring to Cl. 5.3.6 of PDRP-8310-SP-0007 "Design Of Concrete Foundations, Structures & Paving" Factor Of Safety (F.S.) shall be considered as under;

Load Combination description	Minimum Factor of Safety Against Sliding
Long Term : Operating Load without Wind/Seismic Loads	1.50
Short Term : Operating Load + Wind/Seismic Loads	1.50

i) Stability against Uplift

i. Maximum water table shall be considered at Finished Ground Level when foundations are checked for uplift case.

ii. Weight of backfill & effect of soil wedge shall be accounted as counter weight, where it will never be removed. Pits in concrete paved area, which offers considerable soil resistance under normal operating scenario shall be accounted in the design, as complete removal of soil around pit with high water table at grade is never the scenario in routine operating case.

Referring to Cl. 5.3.5 & 5.3.7 of PDRP-8310-SP-0007 "Design Of Concrete Foundations, Structures & Paving" Factor Of Safety (F.S.) shall be considered as under;

Load Combination description	Minimum Factor of Safety Against Uplift
Long Term : Operating Load without Wind/Seismic Loads	1.50
Short Term : Operating Load + Wind/Seismic Loads	1.20
Pits, Channels, Underground Pipes	1.05

j) Loss Of Contact

Allowable loss of contact area between underside of foundation and soil (due to resultant overturning moment) under different loading conditions should be considered as under as far as the overall sliding & overturning criteria are satisfied;

Type of Soil	Operating W/O Wind or Seismic)	Operating + Wind Or Seismic
Foundation on Soil	0 % to 10 %	Up to 25 %

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8.2 Pile Foundations

Pile Foundation system shall be adopted for all critical process plant facilities of EG-ERU (LSTK-1) & Offsite Utility Area (LSTK-2), where in higher bearing capacity are required for 25mm settlement which cannot be satisfied with shallow foundation.

Following are the major list of facilities considered under Pile Foundation;
Process Pipe Rack, Offsite Elevated Pipe Racks & Pipe Bridges, Technological Structure, Compressor Shelter, Compressor Foundation, Tall & Heavy Process Column - Vessel – Drum, Ethylene Storage Sphere, Horizontal Exchanger-Vessel-Drum, and Major Packages like Waste Heat Boiler etc.

Reference shall be made to IPEG-6389-8310-PH-001-0002 "Design Basis For Foundation" for further details and considerations in this regards.
Reference shall be made to IPEG-6389-8320-CA-002-0041 & IPEG-6389-8320-39-002-0012 for Pile Design Calculation & Standard Drawing of Pile.

8.3 Ground Improvement

Various plant & offsite facilities especially Buildings, Cooling Tower, Tankages, as listed in the table below, are having uniform bearing pressure requirement under the foundation footprint are conceptually planned on ground improvement with shallow isolated / isolated raft / combined raft foundations.
Required bearing pressure under footprint of these facilities shall generally be in range of 20 T/Sq.m, for 25mm / 40mm settlement values as applicable depending upon type of foundation.

Ground Improvement by Vibro Stone Column : **(Plant Facilities Required Uniformly Distributed Pressure @ 20 T/Sq.m)**		
1	Buildings : Substation, Control Room, Lab Building, S&D Building, Chemical Storage Building etc.	Shallow Isolated / Raft Footing on Improved Ground
2	Cooling Tower	Basin Raft on Improved Ground
3	Tank Foundation	Ring / Raft foundation on Improved Ground

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Ground Improvement type i.e. stone column or other method & its configuration i.e Dia-Spacing-Square Or Triangle Grid pattern of stone column shall further be conceptualize by LTHE based on FEED stage Geotechnical Investigation Report. Conceptual design shall further be validated by LTHE based on detailed Geotechnical Investigation report & the Prototype Testing of Ground Improvement area prior to actual execution.

Reference shall be made to IPEG-6389-8310-PH-002-0002 "Design Basis For Foundation" for details in this regards.

8.4 **Foundation for Rotating and Reciprocating Equipment**
(Ref. PDRP-8310-SP-0009 Rev. F2 "Design Of Foundations & Structures Supporting Machinery")

Foundation for rotating and reciprocating equipment such as Pump, Blower, Centrifugal Compressor, Reciprocating Compressor, Turbines, Generators etc shall be designed to meet the design (i.e. strength) and performance (i.e. frequency & amplitude) criteria for smooth functioning of rotating equipment.

Foundation shall be Block, Elevated Block or Elevated RC Table Top type as per the requirement specified by the vendor in their GA drawing.

Foundation shall be founded on firm ground either on isolated shallow foundation or pile foundation based on the recommendation given in the Geotechnical Report.

Following are the basic design – performance considerations to be made in design & detailing;

a) Type of Foundation : Open / Pile, Shallow Block / Elevated Block / Elevated Table Top footing shall be decided based on the requirement specified by the vendor

b) Analysis : Equivalent Static & Dynamic Analysis for pump, blower etc shall be done if

i. Machine weighing less than 2.5 Ton Or Power rating less than 300kW Or Mass of the Rotating Parts are less than 1/100th of the mass of foundation, and is installed directly on concrete

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foundation provided that the weight of foundation is not less than 3 times the weight of equipment. For all such cases, only Static Analysis of Foundation is performed.

Necessary design checks like bearing pressure check, RC strength check, eccentricity check, frequency separation check shall be performed by manual calculation (excel base)

ii. Machine weighing more than 2.5 Ton Or Power rating greater than 300kW and mass of its rotating part is greater than 1/100th of the mass of foundation, detail Dynamic Analysis is performed along with other static checks.

Foundation shall be modelled in MAT-3D Or STAAD as (block, elevated RC Table Top Frame Structure) for performing Bearing Pressure Check, Strength Design checks, Frequency Separation & Amplitude check etc.

iii. Irrespective of above criteria, all rotating equipment foundations are checked for frequency separation. It is ensured that the natural frequency of the foundation and machine system is +/- 20% away from operating frequency of machine.

iv. Amplitude of vibration during normal operation or other critical conditions should not exceed the allowable amplitude specified by the equipment manufacturer. In absence of any specific guidelines from vendor, amplitude shall be in conformation to IS: 2974.

v. Bearing and stability check for the foundation is carried out by considering the critical short term cases (Wind or Seismic condition).

vi. Block Foundation has been checked for the following criteria ;

- Top of Block above the finished floor elevation shall be minimum 300 mm above HPP
- Mass Ratio: Wt. of foundation to Wt. of machine shall be 3.0 times for centrifugal machine.
- Mass Ratio: Wt. of foundation to Wt. of machine shall be 5.0 times for reciprocating machine.
- Minimum Reinforcement: As per Cl. No. 5.4.5.1 of IS 2974
- Natural frequency of foundation / structure should not fall within the range of 0.8 to 1.2 of normal operating speed of the equipment.

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- For elevated Table Top foundation, weight of base raft of the frame foundation should not be less than the combined weight of the machine and foundation superstructure above the base raft, i.e. columns, top-deck slab/beams etc.
- The horizontal eccentricity, in any direction, between the center of gravity of the machine - foundation system and the center of base contact area or centroid of the pile group, should be within 5% for block foundations and 3% for frame foundations.
- For frame foundations, base raft should be cast in a single concreting operation. A properly designed construction joint should be provided between the base slab and columns. The entire superstructure of columns and upper deck should be cast in a continuous concreting operation. If height of the frame columns above raft level exceeds 8.0 m, an additional construction joint at the junction of columns / top-deck may be provided.
- The soil stress below foundations under dead loads should not exceed 80% of the allowable soil bearing capacity, or safe load on pile, for static loading.
- All faces of foundation block should be provided with surface reinforcement. Minimum reinforcement diameter in horizontal and vertical directions should be 8mm, and maximum spacing of bars should not exceed 300mm.
- For other details for machine foundation, refer to Doc. No. PDRP-8310-SP-0009.

9. DETAIL DESIGN REQUIREMENTS FOR CONCRETE STRUCTURES

All concrete design shall be conform to limit state method as per IS-456:2000.

8.1 Clear Concrete cover to Reinforcement

(Ref. 5.9 of Doc. No. PDRP-8300-SP-0007 "Design Of Concrete Foundations, Structures & Paving")

Referring to the tender requirement, exposure condition has been considered as "Sever". Accordingly minimum nominal covers to various structural members, are listed in the table below;

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Sr. No.	RCC Element	Clear Cover (mm)
1	Pile	75 mm
	Pile cap : Bottom	100 mm
	Pile cap : Top & Sides	50 mm *
2	Footing : Bottom	50 mm *
	Footing : Sides & Top	50 mm *
	Column, Pedestal & Walls	50 mm *
3	Beam	50 mm *
4	Grade Slab	50 mm
	Elevated Floor Slab	25 mm *
5	contact with liquid / Free Face	50 mm *

(*) or twice the diameter of the reinforcement, whichever is the greater.

Further, requirement of concrete cover / concrete protection shall be defined by LSTK contractor based on the Geotechnical Investigation Report recommendation based on the chemical analysis of soil & ground water.

8.2 **Minimum Thickness of Structural Concrete Elements**

(Ref. 5.9 of Doc. No. PDRP-8300-SP-0007 "Design Of Concrete Foundations, Structures & Paving")

The following minimum thickness shall be followed for various concrete element;

Item Description	Min. Thickness mm
Shallow Foundation	300 mm
Pile Cap	500 mm
Grade Slab	150 mm
Heavy Duty Pavement - for vehicles movement	200 mm
Light Duty Pavement - for general operating area	150 mm
Slab thickness in Raft foundations with beam & slab construction	150 mm
Floor slab / Roof Slab / Walkway / Canopy slab	150 mm
Cable/ Pipe Trench/ Walls & Base Slab	125 mm

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Parapet	100 mm
Precast Trench / UG pits / Manhole Cover	125 mm
Base Slab with beams / without beams	200 / 300 mm thick
Liquid Retaining structure / Underground Pit	
Walls	125 mm
Base slab	125 mm

8.3 **Minimum Reinforcement Bar Diameter & Spacing**

Reference shall be made to Section 4.3 of this document in this regards.

8.4 **Height of Pedestal above HPP / FGL / EGL**

Minimum height of pedestals from Highest Paving Point (HPP) / Finished Ground Level (FGL) / Existing Ground Level (EGL) / Top Of Road etc shall be in line with the following table ;

Item Description	**Pedestal Height (mm)**
Pedestals Typical	300mm above HPP / FGL
Stair / Ladder Pedestals	150~200mm above HPP
Foundation For Equipment, Pump, Packages	300mm min unless otherwise specified in Plot Plan / Mechanical Drawing
Compressors	As per vendor specification
Control Buildings Plinth : - Grade Slab - Raised Floor Area	 HPP + 300mm HPP + 300mm + Height of false flooring
Substation Building Plinth : - Cable Cellar Floor - Transformer Bay - Raised Floor Area	 Top of approach road + 300mm Top of approach road + 150mm with pebbles As per Electrical Requirement
Plant & Other Buildings Plinth	HPP + 300mm

HPP elevation for various units shall be as per Cl.3-d of this document.

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8.5 **Deflection of Concrete members**
(Cl. 6.4 of PDRP-8320-SP-0016)

Horizontal Deflection: Multi-storeyed Steel Structure / Buildings the limiting value of horizontal deflection shall be as under;

Operating Cases (Long Term Case)	: H/325
Loads in combination with Wind	: H/325
Loads in combination with Seismic (Cl. 11.1 of IS1893-Part 4: 2015)	: H/250

Vertical Deflection:

RC Beams (IS 456: 2000)	: Cl. 23.2.1

Separation Gap between Building / Structures:

Separation gap between two RC buildings shall be "(R1Δ1 + R2Δ2)/2" in compliance to Cl.7.11.1 of IS 1893 (Part 1):2016, Amendment No.1, September 2017.

8.6 **Expansion Joint**
(As per Cl. No. 27.2 of IS456: 2000)
For a Reinforcement Concrete Structure expansion joints shall be provided at every 45 m of the plan dimension of the structure, unless a prior approval from PMC/Owner for higher . Expansion joint for building shall be located at every 45m.

10. DETAIL DESIGN REQUIREMENTS FOR STEEL STRUCTURES

9.1 Minimum thickness of various Structural Components (Hot Rolled Sections)

The minimum thickness of various structural components (Hot rolled sections) shall be as given below:

Item Description	Minimum Thickness
Trusses, Bracings, Purlins, and Side girts	6 mm
Columns, Beams	7 mm
Gussets in trusses & girders • up to and including 12 m span • above 12 m span	 8 mm 10 mm

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Item Description	Minimum Thickness
Stiffeners	8 mm
Shear connection plates, End plates	10 mm
Sliding plate under horizontal vessel & Exchanger's base	10 mm
Base plates (plates for moment connection)	10 mm
Chequered plate	6 mm (on plain)
Grating (Electro forged galvanized iron)	25 mm x 5mm

9.2 Deflection of Structural members

Horizontal Deflection: Multi-storeyed Steel Structure / Buildings the limiting value of horizontal deflection shall be as under;

Operating Cases (Long Term Case)	: H/325
Loads in combination with Wind	: H/325
Loads in combination with Seismic (Cl. 11.1 of IS1893-Part 4: 2015)	: H/250

Vertical Deflection:

Beams (Table-6 of IS 800: 2007 Industrial Building)	: L/240
Roof Purlins	: L/200
Gantry girder for supporting dynamic equipment / hoist	: L/450
Gantry girder for manual operated crane	: L/500
Gantry girder for electric overhead crane (Capacity up to 50 T)	: L/750
Gantry girder for electric overhead crane (Capacity > 50 T)	: L/1000
Grating / Chequered Plate (minimum of)	: L/200 or 6mm

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ANNEXURE -1 SITE SPECIFIC SEISMIC SPECTRA

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DBE SEISMIC SPECTRA : Basis For Paradip EG / ERU / OUEP
[Comparision of 5% Damped Spectrum -:- "M/s FUGRO 2008 Study Report" Vs. "IS 1893 (Part-4) : 2015"]

Time Period (s)	Paradip 2008 Spectra (Table 6.2; PDRP-8310-SP-0001)	IS 1893: PART-4 (Annex-B, Fig.1, Hard Strata)			Paradip : LSTK-1 (EG/ERU) & LSTK-2 (OUEP)	Remark
	Spectral Acceleration (g) (Hard Strata) : = (Z/2) * (Sa/g)	Response Acceleration Coefficient (Sa/g)	Zone Factor (Z)	Seismic Spectra = (Z/2) * (Sa/g) (I , R, Df etc to be multiplied as per structure specific consideration)	Seismic Spectra = (Z/2) * (Sa/g)	
0.01	0.11	1.15	0.16	0.09	0.11	Max. of M/s FUGRO's 2008 estimate Vs. IS 1893 is considered by LTHE for design estimate. Above basis shall be validated further w.r.t. Seismic Spectra Study being carried out by LTHE at detail engg. Stage
0.03	0.11	1.45	0.16	0.12	0.12	
0.05	0.12	1.75	0.16	0.14	0.14	
0.07	0.17	2.05	0.16	0.16	0.17	
0.10	0.25	2.50	0.16	0.20	0.25	
0.15	0.32	2.50	0.16	0.20	0.32	
0.20	0.32	2.50	0.16	0.20	0.32	
0.23	0.32	2.50	0.16	0.20	0.32	
0.30	0.32	2.50	0.16	0.20	0.32	
0.39	0.24	2.50	0.16	0.20	0.24	
0.50	0.22	2.00	0.16	0.16	0.22	
0.75	0.16	1.33	0.16	0.11	0.16	
1.00	0.11	1.00	0.16	0.08	0.11	
1.26	0.07	0.79	0.16	0.06	0.07	
1.50	0.06	0.67	0.16	0.05	0.06	
2.00	0.04	0.50	0.16	0.04	0.04	
2.50	0.03	0.40	0.16	0.03	0.03	
3.00	0.02	0.33	0.16	0.03	0.03	
4.00	0.01	0.25	0.16	0.02	0.02	
5.00	0.01	0.20	0.16	0.02	0.02	
6.00	0.01	0.17	0.16	0.01	0.01	
7.00	0.00	0.14	0.16	0.01	0.01	
8.00	0.00	0.13	0.16	0.01	0.01	
9.00	0.00	0.11	0.16	0.01	0.01	
10.00	0.00	0.10	0.16	0.01	0.01	

Steel Structure DBE : 2% Damping			RC Structure DBE : 5% Damping		
Time Period	Damping Multiplicaton Factor	(Z/2) * (Sa/g) * Df	Time Period	Damping Multiplicaton Factor	(Z/2) * (Sa/g) * Df
0.01	1.000	0.110	0.01	1.000	0.110
0.03	1.000	0.116	0.03	1.000	0.116
0.05	1.065	0.149	0.05	1.000	0.140
0.07	1.200	0.206	0.07	1.000	0.172
0.10	1.280	0.320	0.10	1.000	0.250
0.15	1.330	0.426	0.15	1.000	0.320
0.20	1.350	0.432	0.20	1.000	0.320
0.23	1.350	0.432	0.23	1.000	0.320
0.30	1.350	0.432	0.30	1.000	0.320
0.39	1.340	0.322	0.39	1.000	0.240
0.50	1.330	0.288	0.50	1.000	0.217
0.75	1.295	0.211	0.75	1.000	0.163
1.00	1.270	0.140	1.00	1.000	0.110
1.26	1.260	0.088	1.26	1.000	0.070
1.50	1.250	0.075	1.50	1.000	0.060
2.00	1.240	0.050	2.00	1.000	0.040
2.50	1.235	0.040	2.50	1.000	0.032
3.00	1.230	0.033	3.00	1.000	0.027
4.00	1.230	0.025	4.00	1.000	0.020
5.00	1.230	0.020	5.00	1.000	0.016
6.00	1.230	0.016	6.00	1.000	0.013
7.00	1.230	0.014	7.00	1.000	0.011
8.00	1.230	0.012	8.00	1.000	0.010
9.00	1.230	0.011	9.00	1.000	0.009
10.00	1.230	0.010	10.00	1.000	0.008

MCE SEISMIC SPECTRA : Basis For Paradip EG / ERU / OUEP
[Comparision of 5% Damped Spectrum -:- "M/s FUGRO 2008 Study Report" Vs. "IS 1893 (Part-4) : 2015"]

Time Period (s)	Paradip 2008 Spectra (Table 6.3; PDRP-8310-SP-0001)	IS 1893: PART-4 (Annex-B, Fig.1, Hard Strata)			Paradip : LSTK-1 (EG/ERU) & LSTK-2 (OUEP)	Remark
	Spectral Acceleration (g) (Hard Strata) : = (Z/2) * (Sa/g)	Response Acceleration Coefficient (Sa/g)	Zone Factor (Z)	Seismic Spectra = (Z/2) * (Sa/g) (I , R, Df etc to be multiplied as per structure specific consideration)	Seismic Spectra = (Z/2) * (Sa/g)	
0.01	0.22	1.15	0.16	0.18	0.22	Max. of M/s FUGRO's 2008 estimate Vs. IS 1893 is considered by LTHE for design estimate. Above basis shall be validated further w.r.t. Seismic Spectra Study being carried out by LTHE at detail engg. Stage
0.03	0.22	1.45	0.16	0.23	0.23	
0.06	0.26	1.90	0.16	0.30	0.30	
0.07	0.31	2.05	0.16	0.33	0.33	
0.10	0.44	2.50	0.16	0.40	0.44	
0.16	0.60	2.50	0.16	0.40	0.60	
0.20	0.60	2.50	0.16	0.40	0.60	
0.23	0.60	2.50	0.16	0.40	0.60	
0.31	0.60	2.50	0.16	0.40	0.60	
0.39	0.52	2.50	0.16	0.40	0.52	
0.50	0.47	2.00	0.16	0.32	0.47	
0.75	0.37	1.33	0.16	0.21	0.37	
1.00	0.26	1.00	0.16	0.16	0.26	
1.26	0.17	0.79	0.16	0.13	0.17	
1.50	0.14	0.67	0.16	0.11	0.14	
2.00	0.08	0.50	0.16	0.08	0.08	
2.50	0.06	0.40	0.16	0.06	0.06	
3.00	0.04	0.33	0.16	0.05	0.05	
4.00	0.03	0.25	0.16	0.04	0.04	
5.00	0.02	0.20	0.16	0.03	0.03	
6.00	0.01	0.17	0.16	0.03	0.03	
7.00	0.01	0.14	0.16	0.02	0.02	
8.00	0.01	0.13	0.16	0.02	0.02	
9.00	0.01	0.11	0.16	0.02	0.02	
10.00	0.00	0.10	0.16	0.02	0.02	

Steel Structure MCE : 4% Damping			RC Structure MCE : 7% Damping		
Time Period	Damping Multiplicaton Factor	(Z/2) * (Sa/g) * Df	Time Period	Damping Multiplicaton Factor	(Z/2) * (Sa/g) * Df
0.01	1.000	0.220	0.01	1.000	0.220
0.03	1.000	0.232	0.03	1.000	0.232
0.06	1.016	0.309	0.06	0.975	0.296
0.07	1.049	0.344	0.07	0.939	0.308
0.10	1.068	0.470	0.10	0.913	0.402
0.16	1.080	0.648	0.16	0.896	0.538
0.20	1.085	0.651	0.20	0.897	0.538
0.23	1.085	0.651	0.23	0.897	0.538
0.31	1.085	0.651	0.31	0.897	0.538
0.39	1.063	0.563	0.39	0.893	0.464
0.50	1.080	0.511	0.50	0.896	0.424
0.75	1.072	0.393	0.75	0.900	0.330
1.00	1.066	0.277	1.00	0.903	0.235
1.26	1.061	0.180	1.26	0.903	0.154
1.50	1.061	0.149	1.50	0.903	0.127
2.00	1.059	0.085	2.00	0.903	0.072
2.50	1.058	0.068	2.50	0.903	0.058
3.00	1.056	0.056	3.00	0.903	0.048
4.00	1.056	0.042	4.00	0.903	0.036
5.00	1.056	0.034	5.00	0.903	0.029
6.00	1.056	0.028	6.00	0.903	0.024
7.00	1.056	0.024	7.00	0.903	0.021
8.00	1.056	0.021	8.00	0.903	0.018
9.00	1.056	0.019	9.00	0.903	0.016
10.00	1.056	0.017	10.00	0.903	0.014

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Table 6-2. 5-Percent Damped Horizontal DBE Spectrum

Period (s)	Spectral Acceleration (g)
0.01	0.11
0.03	0.11
0.05	0.12
0.10	0.25
0.15	0.32
0.23	0.32
0.39	0.24
1.00	0.11
1.26	0.07
2.0	0.04
2.5	0.03
3.0	0.02
4.0	0.01
5.0	0.01
6.0	0.01
7.0	0.00
8.0	0.00
9.0	0.00
10.0	0.00

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Table 6-3. 5-Percent Damped Horizontal MCE Spectrum

Period (s)	Spectral Acceleration (g)
0.01	0.22
0.03	0.22
0.06	0.26
0.10	0.44
0.16	0.60
0.31	0.60
0.39	0.52
1.0	0.26
1.26	0.17
2.00	0.08
2.5	0.06
3.0	0.04
4.0	0.03
5.0	0.02
6.0	0.01
7.0	0.01
8.0	0.01
9.0	0.01
10.0	0.00

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APPENDIX : EXTRACT FROM FUGRO REPORT NO. C696-04 PART 2

Table 6-4. Modification Factors to Obtain Spectra for Other Damping Ratios

Period (s)	Damping Ratio					
	0.5%	2%	3%	4%	7%	10%
0.01	1.000	1.000	1.000	1.000	1.000	1.000
0.03	1.000	1.000	1.000	1.000	1.000	1.000
0.05	1.163	1.065	1.036	1.016	0.985	0.967
0.075	1.503	1.200	1.111	1.049	0.939	0.875
0.1	1.704	1.280	1.156	1.068	0.913	0.820
0.15	1.829	1.330	1.184	1.080	0.896	0.785
0.2	1.880	1.350	1.195	1.085	0.897	0.779
0.25	1.880	1.350	1.195	1.085	0.897	0.779
0.3	1.880	1.350	1.195	1.085	0.897	0.779
0.35	1.867	1.345	1.192	1.084	0.891	0.775
0.4	1.854	1.340	1.190	1.083	0.893	0.780
0.5	1.829	1.330	1.184	1.080	0.896	0.785
0.6	1.779	1.310	1.173	1.076	0.898	0.790
0.7	1.754	1.300	1.167	1.073	0.898	0.790
0.8	1.729	1.290	1.162	1.071	0.900	0.795
0.9	1.704	1.280	1.156	1.068	0.902	0.798
1	1.678	1.270	1.150	1.066	0.903	0.800
1.5	1.628	1.250	1.139	1.061	0.903	0.800
2	1.603	1.240	1.134	1.059	0.903	0.800
3	1.578	1.230	1.128	1.056	0.903	0.800
4	1.578	1.230	1.128	1.056	0.903	0.800
≥5	1.578	1.230	1.128	1.056	0.903	0.800

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4.2.2 Empirical Attenuation Relationships

The attenuation of seismic waves from a seismogenic source were modeled using empirical attenuation functions. These empirical functions should model the type of rupture mechanism as well as the regional geology to properly estimate site-specific strong ground motion.

The project site lies at the edge of the stable continental region of the Indian subcontinent and the extended crustal region. For this study the assumption was made that shallow crustal earthquakes in and around the Paradip Refinery site are similar to comparable events in the relatively stable areas of the Eastern United States. Additionally, selection of attenuation relationships reflected the presence of the project site within an extended crust environment.

For the stable continental regions, two equally weighted attenuation relationships were used; Atkinson and Boore (1997) and Toro et al. (1997). For the extended crust areas, the attenuation relationship developed by Toro et al. (1997) for the extended crust environment of the gulf crustal regions was used in conjunction with the Eastern Continental relationships mentioned above. The Toro et al. (1997) gulf relationship was given a weight of 0.50 while the other relationships were given weights of 0.25 each. The various attenuation relationships and their respective weights are presented in Table 4-2.

Table 4.2. Horizontal Ground Motion Attenuation Relationships

Seismic Source	Attenuation Equation	Weight
All Seismic Sources except Extended Crust Sources	Atkinson and Boore (1997)	0.5
	Toro et al. (1997) – Eastern and Continental US	0.5
Extended Crust Zones	Atkinson and Boore (1997)	0.25
	Toro et al. (1997) – Eastern and Continental US	0.25
	Toro et al. (1997) – Gulf Crustal Region	0.5

The subsurface information at the project site is available to a maximum depth of 30 meters. Based on our understanding of the regional geology in the project area, competent bedrock is anticipated to be present at significant depths from the ground surface. Therefore, results were developed for a "hard soil" horizon, with an average shear wave velocity in the upper 30 meters ($V_{S,30}$) of 500 m/s. Atkinson and Boore (1997) have developed attenuation relationship for soils with $V_{S,30}$ of 500 m/s as well as for competent bedrock. However, the Toro et al. (1997) relationships were developed only for bedrock. Therefore, spectral amplification factors were obtained by comparing the Atkinson and Boore (1997) bedrock and "hard soil" results, and were subsequently applied to the results from the Toro et al. relationships in order to obtain equivalent results applicable at the "hard soil" horizon.

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4.2.6 Extending the Design Spectra to Long Periods

Due to the scarcity of well-recorded long-period ground motions, empirical attenuation relationships used for this project have traditionally been developed for structural periods up to 2 seconds. As a result, the design spectra from probabilistic seismic hazard analyses are well defined up to periods of approximately 2 seconds. For the proposed tanks being considered for this project, the sloshing periods may be longer. Consequently, design data were extended to longer structural periods than those for which empirical attenuation relationships are available.

A well-established methodology to extend a design spectrum to longer periods is not available in the literature. However, there appears to be consensus among the scientific community (e.g., Silva and Abrahamson, 1992; NEHRP, 2003) that in the absence of data at long periods design response spectra can be extended based on a constant spectral velocity assumption for the intermediate periods and a constant spectral displacement assumption for long periods. For design purposes the "corner period", marking the transition between constant spectral velocity and constant spectral displacement is a function of the earthquake magnitude and has been tabulated by NEHRP (2003). For this study, the corner period was estimated to be approximately 6 to 8 seconds for various return period events, corresponding to contributing magnitudes ranging from 7.0 to 8.0. Between structural periods of 8 and 10 seconds, the target spectra were extended based on an assumption of constant spectral displacement. Between structural periods of 2 and 8 seconds the target spectra were interpolated using a smooth curve on a tripartite plot.

4.3 DEFINITION OF DESIGN LEVEL EVENTS

Based on the information from Foster Wheeler, the various facilities for the Paradip Refinery project will be designed in accordance with the Indian Standard IS 1893 (Part 4), *Criteria for Earthquake Resistant Design of Structures*. According to IS 1893, two levels of design ground motions are developed:

1. DBE (Design Basis Earthquake) and
2. MCE (Maximum Considered Earthquake).

The DBE is defined per IS 1893 as an event with 5 percent probability of exceedance in 50 years (i.e., a return period of 475 years). The MCE is defined as an event with 2 percent probability of exceedance in 50 years (i.e., a return period of 2,475 years). Additionally, hard soil ground motions were with a 1 percent probability of exceedance in 50 years (i.e., a return period of 4,975 years) were also estimated per the request of Foster Wheeler.

4.4 RESULTS

Results from the seismic hazard analyses are presented in terms of design response spectra for horizontal and vertical ground motions. As described in Section 4.2.2, the results are based on attenuation relationships for a "hard soil" site classification, i.e., shear wave velocity on the order of 500 meters per second.

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